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# Evaluation of the impact of Urban Stream Syndrome on macroinvertebrate functional feeding diversity and organic matter processing in streams

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# Evaluation of the impact of Urban Stream Syndrome on macroinvertebrate functional feeding diversity and organic matter processing in streams

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Honors in Biology

Committee members: Dr. Dina Leech (Longwood University), Dr. Scott Star (Hampden-Sydney College), Dr. Brian Brown (Virginia Polytechnic Institute and State University).

# Evaluation of the impact of Urban Stream Syndrome on macroinvertebrate functional feeding diversity and organic matter processing in streams

by

Helena Loucas

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## Abstract

Urbanization causes the degradation of streams described as the Urban Stream Syndrome, resulting in increased contamination, altered stream morphology, and reduced taxonomic richness. Macroinvertebrates vary in their tolerance to the impacts of urbanization, therefore the reduction of taxonomic richness increases the abundance of tolerant species and decreases the abundance of intolerant species. Aquatic macroinvertebrates are specialized feeders, falling into five functional feeding groups which affect ecosystem processes such as the decomposition of organic matter. Through a reduction of intolerant taxa, functional feeding diversity will be reduced because the functional feeding groups are not equally distributed across all tolerance levels. This alteration in functional feeding diversity may affect ecosystem processes within the stream. I hypothesized that the degradation of streams from the Urban Stream Syndrome would result in a decrease in macroinvertebrate diversity and the diversity of the functional feeding groups, altering organic matter processing. To evaluate this hypothesis, I quantified the taxonomic and functional diversity of macroinvertebrate assemblages in an urban and non-urban stream. I then developed a model for organic matter processing using macroinvertebrate consumption rates from the literature and predicted organic matter processing rates in the urban and non-urban stream. I found that there was a reduction in the intolerant macroinvertebrates in the urban stream when compared to the non-urban stream. There was also a reduction in taxonomic and functional diversity in the urban stream. The modeled consumption rates indicated that in the urban stream, Collectors were responsible for the most consumption, whereas in the non-urban stream Shredders were predicted to process the most organic matter. These results support my hypothesis that the Urban Stream Syndrome alters organic matter processing in streams through a shift in macroinvertebrate functional diversity.

# Dedication

I would like to dedicate this thesis to my family, who are the reason I am at Longwood University. Additionally, I'd like to dedicate it to my faculty advisor, Dr. Fortino, for all of his help throughout my undergraduate experience. Finally, I'd like to dedicate this to my friend, Grace Smith, who spent many hours by my side while I worked on this project.

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## Introduction

**Impacts of urbanization on stream ecosystems:** Increases in urban land use put greater pressure on surrounding watersheds than non-urban land use (Meyer et al., 2005). Urbanization causes the degradation of streams in urbanized watersheds in a process known as the Urban Stream Syndrome which is characterized by an increase in concentrations of contaminants, altered stream morphology, altered physical processing of organic matter, and reduced taxonomic richness (Walsh et al., 2005, Chadwick et al., 2006, Komínková 2012; Figure 1). Symptoms associated with the change in environmental conditions due to the Urban Stream Syndrome, such as increased nutrient concentrations and increasing concentration of suspended matter, result in oxygen depletion and altered redox reactions (Komínková 2012). The expansion of urbanization has caused altered water quality through changes in pH, dissolved oxygen content, and total suspended solids (Ferreira et al., 2021), changes in light penetration (Reisinger et al., 2019), a simplification of stream substrate and deepening of the stream channel (Violin et al., 2011), and reduced nutrient uptake by the stream community (Meyer et al., 2005). Furthermore, increases in impervious surfaces have been shown to be correlated with reductions in biodiversity across ecoregions (Barnum et al., 2018). The impact of urbanization on stream systems occurs across multiple levels of stream structure and function. These compound changes in environmental conditions have impacts at the community level because stream communities, including macroinvertebrates, are influenced by the environmental templet (Townsend and Hildrew 1994, Wallace and Webster 1996).

**Effects of Urban Stream Syndrome on Macroinvertebrates:** The macroinvertebrate communities in urban and non-urban streams differ mainly due to macroinvertebrate sensitivity to stream abiotic conditions. Stream environmental conditions like PH, dissolved oxygen content, and stream substrate affect the taxonomic composition of the system, because environmental conditions influence which taxa can and cannot survive there (Wallace and Webster 1996, Barnum et al., 2018, Keinath et al., 2023). Stream macroinvertebrates can be classified based on their ability to tolerate accumulated anthropogenic stressors, and reductions in tolerant taxa have been used to assess stream biotic integrity (Hilsenhoff 1987 and Lenat 1993). The impact of alterations to environmental conditions is sufficient that Bressler et al., (2006) was able to predict the presence and absence of stream macroinvertebrate groups across a gradient of known anthropogenic stressors. In cases of increased urban impact, changes in the assemblages of macroinvertebrates occur predictably as a decrease in intolerant species (Tampo et al., 2021, Wan Abdul Ghani et al., 2018). Ergo, the taxonomic composition of an anthropogenically impacted system is driven primarily by the abiotic conditions, and tolerant species more likely to be present due to their ability to survive in the poor conditions (Lenat 1993). In systems impacted by Urban Stream Syndrome, macroinvertebrate diversity is decreased and species of macroinvertebrates which are incapable of tolerating these poor conditions will decline. Therefore, the alterations in environmental conditions due to urbanization are associated with a reduction of biotic richness, accompanied by an increase in dominance of more tolerant species of macroinvertebrates (Meyer et al., 2005, Walsh et al., 2005; Figure 1).

**Effects of Urban Stream Syndrome on Macroinvertebrate Functional Diversity:** Changes in the taxonomic diversity of a system affects the functional diversity of the system through changes in organismal traits (Meyer et al., 2005, Poff et al., 2006). Functional traits can be conceptualized as either response traits, which affect an organism's fitness or effect traits, which affect an organism's impact on the systems in which it occurs (Violle et al., 2007). Because effect traits are linked to response traits in individual organisms, the function that an organism performs in a system will be connected to its ability to tolerate the specific environmental conditions of the system (Violle et al., 2007). Therefore, in streams, there is a relationship between macroinvertebrate taxonomic diversity, functional diversity, and environmental processes (Sun et al., 2021) operating on the connection between response traits and effect traits (Violle et al., 2007). This relationship between functional diversity and stream environmental processes manifests significantly through macroinvertebrate feeding behaviors (Guo et al., 2018). Macroinvertebrate feeding contributes to numerous environmental processes, primarily through the decomposition of standing stocks of organic matter (Newman 1991). Aquatic macroinvertebrates meet their nutritional needs by consuming the organic nutrients in detrital or living organic matter (Wallace and Webster 1996) and thus possess specialized traits to increase feeding efficiency. Although, not all macroinvertebrates can be classified into functional feeding groups due to omnivory, many aquatic macroinvertebrates are specialized feeders, falling into five functional feeding groups (Scrapers, Shredders, Collectors, Filterers, and Predators) with different ecological niches (Merritt et al., 2017, Vannote et al., 1980). The diversity of functional feeding groups represents, in part, the functional diversity of the system, which includes traits that influence ecosystem processes (Tilman 2001, Sun et al., 2021). Therefore, variation in taxonomic composition should exert a strong influence on environmental processes because the

species present occupy different niches and possess specific effect traits that determine ecosystem function (Figure 1).

## Research Question and Hypothesis

This study evaluated the connection between the observed reduction of macroinvertebrate richness due to the changes associated with the Urban Stream Syndrome and how this reduction in richness may alter functional diversity and ecosystem processes associated with organic matter cycling in urban streams (Figure 1). Specifically, my study sought to address the question: *How do changes in macroinvertebrate functional diversity due to urbanization alter organic matter processing?*

I hypothesized that: *The altered environmental conditions associated with Urban Stream Syndrome will reduce the presence of intolerant macroinvertebrate taxa, ultimately altering environmental processes via reduction in functional feeding diversity.*

The lack of intolerant species present in urban systems was hypothesized to result in a decrease in the decomposition of organic matter due to the reduction in functional diversity to only a few functional feeding groups dominating the community (Tampo et al., 2021). Since, functional feeding groups represented by intolerant species would not be present in streams impacted by the Urban Stream Syndrome, these streams would have a decreased rate of organic matter processing.

I evaluated my hypothesis with three research objectives. For my first objective I compare the macroinvertebrate taxonomic and functional diversity of an urban and non-urban stream. The results of the first objective were then used complete the second objective, which was to model organic matter processing based on observed functional diversity in urban and non-

urban streams. Finally, my third objective was to compare the environmental conditions of the urban and non-urban stream to confirm that the conditions are aligned with those commonly seen in streams suffering from the Urban Stream Syndrome.

## Methods

In order to compare macroinvertebrate functional diversity in an urban and non-urban stream, I selected three replicate stream reaches in Gross Creek (37.304, -78.389), a second-order stream with an almost entirely urban watershed and three replicate stream reaches in Hudson's Branch (37.246, -78.473), a second-order stream with an almost entirely forested watershed. The watershed land use was visually estimated with the 1:24,000 USGS maps for Hampden Sydney Virginia and Farmville Virginia. Each replicate stream reach contained a riffle, run, and pool (hereafter "habitat"). Both streams were sampled on September 10$^{th}$, 2024. For each habitat in each replicate reach I used a meter tape to measure the channel width in meters across the approximate center of the habitat patch and measured the maximum depth in centimeters at three points in the approximate center of the of the habitat patch before averaging them. Using a meter tape, I also measured the right and left bank heights by taking the measurement from the highest point on the bank of the habitat. The high-water mark was estimated from the location of organic debris or trash along the bank or in trees. I also described the substrate of each habitat in each reach by visually estimating the percent coverage of sand, cobble, boulder, or bedrock substrate at each habitat plot. Sand was classified as any sediment less than approximately1 mm in diameter, pebble was classified as any sediment between 1 mm and 1 cm in diameter, cobble was classified as any sediment between 1 cm and 10 cm in diameter, and bedrock was classified as immovable rock substrate. I measured canopy cover of each habitat patch using a densiometer.

To quantify the macroinvertebrate community in each stream, I used a Hess Sampler with a 0.0006 $m^2$ sample area and 250 µm mesh netting to collect one sample of macroinvertebrates from each replicate habitat patch within each creek. All collected macroinvertebrates were preserved in 70 % ethanol in the field and transported back to the lab for processing.

I separated the macroinvertebrates from substrate material with a dissecting microscope and then counted and identified each organism to the lowest taxonomic level possible using Merrett, Cummins, and Berg (2008). There were 7 of the total 18 samples that I collected that contained too much substrate material to efficiently sort and therefore these samples were split once with a Folsom Plankton Splitter prior to sorting, identifying, and counting. The final counts from the split samples were multiplied by 2 to account for the splitting.

Each taxon was assigned to a functional feeding group using an online visual identification key (Macroinveretbrates.org) and a tolerance value using the "Purdue Extension Bioindicators of Water Quality Quick Reference Guide" (Speelman and Carroll 2016). I normalized the abundance of each taxon and the taxa in each functional group to density (individuals $m^{-2}$). I used the vegan package (Oksanen et al., 2024) to calculate both the Inverse Simpson (Simpson, 1949) and Shannon-Weaver index (Shannon and Weaver, 1949) diversity indices for both the taxonomic and functional diversity of each sample. Differences in the density, richness, taxonomic diversity, functional diversity, mean tolerance, and abundance-weighted tolerance between streams and habitats were tested using a 2-way ANOVA with habitat nested within stream. I assessed macroinvertebrate tolerance using both the un-weighted measurements and an abundance-weighted average of the tolerance levels to account for the abundance of high tolerance taxonomic groups in all of the samples (Table 1).

I ran an NMDS ordination to simplify the characterization of the sites based on community composition and function. I used the "metaMDS" function in the vegan package in R to run an ordination characterizing the sites based on the taxonomic diversity and the functional diversity. I then assessed the sorting of the sites in multivariate space graphically by plotting the sample loadings on the first axis of variation (MDS1) by site for taxonomic diversity and functional diversity, respectively. The distance metric of the NMDS ordination was Bray Curtis and the transformation was a square root. I used 2 dimensions because the Shepards diagram showed good fit ($R^2 > 0.9$) between the reduced and observed dissimilarity with 2 dimensions.

To evaluate the effect of changes in functional diversity on ecosystem processes, organic matter processing was modeled based on observed functional diversity in urban and non-urban streams. The model was constructed using the data on the organic matter consumption rate (g AFDM $m^{-2}$ $y^{-1}$) of macroinvertebrate functional feeding groups in Stagliano and Whiles (2002). These consumption rates were collected in the Spring and Fall of 1998 (Stagliano and Whiles 2002). The consumption rates in Stagliano and Whiles (2002) were reported per square meter but were converted to per individual using the abundance data also provided. These estimates of organic matter consumption per individual were then multiplied by the observed density of each macroinvertebrate functional feeding group for each site (Gross Creek and Hudson's Branch) and habitat (riffle, run, and pool) to get an estimate of the organic matter consumption rate (g AFDM $m^{-2}$ $y^{-1}$) in each habitat in the urban and non-urban stream.

All data manipulation and analyses were performed in R version 4.4.0 (R Core Team, 2024).

## Results

### Objective 1: Compare macroinvertebrate functional diversity in an urban and non-urban stream

In order to understand how urbanization alters the macroinvertebrate diversity of streams, I compared the macroinvertebrate communities of Gross Creek and Hudson's Branch, an urban and non-urban stream, respectively (Table 1, Figure 2). The mean (± 1 SD) macroinvertebrate density of Hudson's Branch was 1654 (± 1059) individuals $m^{-2}$, which was not significantly different from the macroinvertebrate density of 2318 (± 1592) individuals $m^{-2}$ found in Gross Creek, ($p = 0.343$). Similarly, there was no significant difference in the mean (± 1 SD) macroinvertebrate densities of 1560 (±1193), 2383 (±1827), 2014 (±1038) measured in the riffles, runs, and pools, respectively ($p = 0.683$). In contrast to density, mean taxa richness did differ between the sites ($p < 0.001$). I found a mean (± 1 SD) of 7.00 (± 2.83) taxa in Hudson's Branch of but only 2.78 (± 0.833) taxa in Gross Creek. Habitat, however, had no effect on macroinvertebrate richness ($p = 0.163$).

While there was overall agreement between the Inverse Simpson and Shannon-Weaver diversity indices that Hudson's Branch had greater taxonomic diversity, there were some differences in the impact of habitat (Table 2, Figure 3, Figure 4). The Inverse Simpson taxonomic diversity was found to be significantly greater in Hudson's Branch than in Gross Creek ($p = 0.003$; Table 3). The Inverse Simpson taxonomic diversity also differed among the habitats ($p = 0.023$; Table 3). The Shannon-Weaver taxonomic diversity was found to be significantly greater in Hudson's Branch than in Gross Creek ($p = 0.001$), however there was

only a marginally significant difference in the Shannon-Weaver taxonomic diversity among the habitats (p = 0.062; Table 3).

The taxa collected from the streams were assigned to one of 5 functional feeding groups (Appendix 1) allowing me to measure functional differences between the streams and habitats. Both the Inverse Simpson and Shannon-Weaver diversity indices show that the functional diversity was significantly greater in Hudson's Branch, than in Gross Creek, however, unlike taxonomic diversity, there were no differences in the impact of habitat on functional diversity (Table 3, Figure 5, Figure 6). The Inverse Simpson functional diversity of Hudson's Branch was found to be significantly greater than that of Gross Creek (p= 0.016; Table 3). However, there was no significant difference in the Inverse Simpson Functional diversity among the habitats (p = 0.161; Table 3). The Shannon functional diversity of Hudson's Branch was found to be significantly greater than that of Gross Creek (p = 0.003; Table 3). However, there was no significant difference in the mean Shannon functional diversity among the habitats (p = 0.192; Table 3).

The NMDS ordinations using macroinvertebrate taxonomic identity (Figure 7) and functional feeding groups (Figure 8) produced final stresses of 0.0988 and 0.0984, respectively. For both ordinations the urban and non-urban streams showed separation along the first axis of variation (NMDS1).

The taxonomic ordination showed that Gross Creek sites had a mean loading of 0.528 (+/- 0.217) on NMDS1 with a minimum of 0.188 and a maximum of 0.777, in contrast the Hudson's Branch sites had a mean loading of -0.528 (+/- 0.697) with a minimum of –1.34 and a maximum of 0.425 (Figure 7, Figure 9). For the taxonomic ordination in Hudson's Branch the riffles and runs with mean (+/- 1SD) loading values of -0.989 (+/- 0.400) and -0.837 (+/- 0.542),

respectively, separated from the pools, which had a mean (+/- 1SD) loading of 0.243 (+/- 0.193), which was more similar to all the Gross Creek sites (Figure 7, Figure 9). In Gross Creek all three habitats loaded similarly to each other on the taxonomic NMDS1 with an overall mean (+/- 1SD) of 0.528 (+/- 0.217) and were distinct from the riffles and runs of Hudson's Branch (Figure 9). The functional ordination showed that Gross Creek sites had a mean loading of –0.364 (+/- 0.167) on NMDS1 with a minimum of –0.542 and a maximum of –0.0676, in contrast the Hudson's Branch sites had a mean loading of .364 (+/0.342) with a minimum of –0.242 and a maximum of 0.843 (Figure 8, Figure 10). For the functional ordination in Hudson's Branch the riffles and runs with mean (+/- 1SD) loading values of .635 (+/- 0.239) and .361 (+/- 0.327), respectively, slightly separated from the pools, which had a mean (+/- 1SD) loading of .096 (+/- 0.293). However, they were not as noticeably separate as in the taxonomic ordination (Figure 8, Figure 10). In Gross Creek all three habitats loaded similarly to each other on the functional NMDS1 with an overall mean (+/- 1SD) of -0.364 (+/- 0.167) and were distinct from the habitats of Hudson's Branch (Figure 10).

In Hudson's Branch the un-weighted tolerances of the macroinvertebrates I collected ranged from 0 to 8, with a median value of 5. In contrast, in Gross Creek the un-weighted tolerances of the collected macroinvertebrates ranged only from 5 to 8, with a higher median value of 8 (Table 4, Figure 11). The specific habitats at each site, largely followed the patterns of the site overall. In all habitats at both sites, the maximum tolerance observed was always 8, because chironomids (the most tolerant macroinvertebrates collected) were present in all samples (Table 1). In Hudson's Branch, the minimum tolerance was 0 for pools and 1 for riffles and runs, but these values were not contributed by the same taxa. In pools the minimum tolerance values were due to *Corydalidae nigronia*, whereas, *Capniidae allocapnia* were the least tolerant taxon

found in runs. In riffles, the lowest tolerance taxa were *Capniidae allocapnia*, *Lepidostomidae lepidostoma*, *Chloroperidae alloperla*, and *Perlidae paragnetina* (Table 1). Reflecting the lower diversity, the least tolerant organism found in all habitats in Gross Creek was always the larvae in the genus *Anchytarsus* (Table 1).

Weighting the mean tolerance of each sample by the abundance of each taxon reflected the disproportionate representation of certain taxa in the system, specifically the chironomids. Since the chironomids were overwhelmingly the most abundant taxa collected from both streams and among the most tolerant (Table 1), the abundance-weighted means were more similar between the sites (Table 5, Figure 12), although the mean (± 1 SD) abundance-weighted tolerance of 7.21 (± 0.624) for Hudson's Branch was significantly less than ($p < 0.001$) Gross Creek, with the mean (± 1 SD) abundance-weighted tolerance of 7.96 (± .104). I found a significant difference in abundance-weighted tolerance amongst the habitats as well ($p = 0.009$; Table 5, Figure 12).

However, the abundance-weighted tolerance differed amongst the habitats with the mean (± 1 SD) values of pool habitats being 7.68 (± 0.210), for Hudson's Branch and 7.99 (± 0.011) for Gross Creek. The mean (± 1 SD) values for riffle habitats were 6.56 (± 0.543) for Hudson's Branch and 7.89 (± 0.178) for Gross Creek. The mean (± 1 SD) values for run habitats were and 7.41 (± 0.437) for Hudson's Creek and 7.99 (± 0.010) for Gross Creek (Table 5).

**Research Objective 2: Model organic matter processing based on observed functional diversity in urban and non-urban streams.**

I used literature values for the organic matter consumption rate of macroinvertebrates to model how the observed densities of macroinvertebrates in Gross Creek and Hudson's Branch would affect the processing of organic matter in these streams. The results of the model indicate

that organic matter consumption at each site is variable at the site scale with standard deviations typically of the same order of magnitude as the means, and reflecting the variation in the observed density of the macroinvertebrates (Table 6). In Gross Creek, the model predicted that the greatest mean organic matter consumption rate would be due to Collectors, in which the predicted mean consumption was 3 times greater than the next highest predicted mean consumption by the Predators (Table 6). In Hudson's branch, the model predicted that the greatest mean organic matter processing would be due to Shredders, which were predicted to be 5.4 times greater than the next highest mean consumption rate by the Collectors (Table 6). Scrapers were predicted to have the lowest contribution to organic matter consumption in both streams (Table 6). The biggest difference in the predicted organic matter consumption rate between the sites was due to Shredders, whose predicted organic matter consumption was 28 times greater in Hudson's Branch than in Gross Creek, and due to Filterers, who's predicted organic matter consumption rate was 15 times greater in Hudson's Branch than Gross Creek (Table 6). In Gross Creek, the organic matter consumption rate of Collectors and Predators was predicted to be greater than in Hudson's Branch but the differences were more modest at 1.8 and 1.7 times, respectively (Table 6).

In Hudson, the mean organic matter processing by Collectors was fairly uniform across habitats. Mean organic matter processing by Filterers was greatest in the Riffles but was also most variable there (Table 7). Mean organic matter processing by Predators was fairly uniform across habitats with the most variation in the pools. Organic matter processing by Scrapers was only found in Riffles and Runs and more variable in the Riffles. Organic matter processing by Shredders was about 3 times greater in Riffles than the Pools or Runs but was least variable in the Runs.

In Gross, the mean organic matter processing by Collectors was not uniform across habitats, with most taxa found in Pool and Run habitats (Table 7). Mean organic matter processing by Predators was fairly uniform across habitats with the most variation in the Riffles. Organic matter processing by Shredders was about greater in Runs than the Pools and Riffles. Mean organic matter processing by Filterers was almost nonexistent, with a small presence in Riffle habitats and no presence in Pools and Runs (Table 7). Similarly, scrapers were completely unrepresented in all habitats of the system.

**Research Objective 3: Quantify environmental conditions in urban and non-urban streams.**

Overall, the sampled reaches of Gross Creek were wider than the sampled reaches of Hudson's Branch, with the exception of the riffles. The mean width of the riffle habitats in Gross Creek was 2.23 meters, that was similar to Hudson's Branch, which had a mean riffle width of 2.63 meters. Both the pool and run habitats of Gross Creek were wider than those in Hudson's Branch (Table 8). The mean width of the pool habitats in Gross Creek was 3.7 meters, differing from the mean width of 1.7 meters for the pool habitats in Hudson's Branch. The mean width of the run habitats in Gross Creek was 3.23 meters, whereas the mean width of the run habitats in Hudson's Branch was 2 meters. The mean depths of the habitats in Gross Creek were all deeper than those of Hudson's Branch (Table 8). The mean depth of the riffle habitats was 7.56 cm for Gross Creek, with Hudson's Branch being a shallower 4.22 cm. The difference in mean depths of the pool habitats was more varied with 41.67 cm for Gross Creek and 33.89 cm for Hudson's Branch. The mean depth of the run habitats was 14 cm for Gross Creek and a shallower 9.67 cm for Hudson's Branch.

The riparian land-use around the two sites also differed. The vegetation surrounding the riffle, run, and pool habitats of Gross Creek were primarily made up of trees, shrubs, and short grasses. The vegetation surrounding the riffle, run, and pool habitats of Hudson's Branch were primarily made up of forested area.

The sediments within the riffle habitats of Gross Creek were primarily sand and cobble, whereas the sediments within the riffle habitats of Hudson's Branch were primarily cobble and bedrock. The sediments within the pool habitats of Gross Creek were primarily sand, and the sediments within the pool habitats of Hudson's Branch were primarily sand and cobble. The sediments within the run habitats of Gross Creek were primarily sand and cobble, while the sediments within the run habitats of Hudson's Branch were a mix of pebbles, sand, and cobble.

## Discussion

The results of my study support the relationship between stream macroinvertebrate taxonomic diversity, functional diversity, and environmental processes (Sun et al., 2021). I found that Gross Creek, which drains a predominantly urban watershed differed in its environmental parameters, taxonomic diversity, and functional diversity from Hudson's Branch, a stream with a predominantly forested watershed. Furthermore, my model of macroinvertebrate organic matter consumption rates predicted that there would be a shift away from shredding and toward collecting in the urban stream.

The physical differences between Hudson's Branch and Gross Creek were mainly associated with surrounding land-use and bank incision. Both creeks are similar in size and most environmental parameters but the surrounding land use of Gross Creek is almost entirely urban, as it runs parallel to Main Street in the town of Farmville, VA. Hudson's Branch, in contrast, is

surrounded by forested land use. Despite its predominantly forested watershed, Hudson's Branch is not entirely unaffected by anthropogenic stressors. The creek's watershed contains some development associated with its close proximity to Hampden Sydney College's campus and like most creeks in the region is likely still affected by the long history of agricultural land use in central Virginia, which alters stream community composition, structure, and function (Dance & Hynes, 1980). Gross Creek has several characteristics that align with the Urban Stream Syndrome, particularly notable in the increased bank height and incision, which differed from the more consistent banks of Hudson's Branch (Walsh et al., 2005). Also of note is the difference in riparian vegetation in relation to watershed land use. Gross Creek's surrounding vegetation was primarily shrubbery and grasses with some trees, and no trees in areas closer to roadway bridges, whereas Hudson's Branch was entirely surrounded by forest. Although not quantified in this study, my observation of considerable industrial construction fill (e.g., concrete rubble) and human trash within Gross creek was also consistent with what is seen in urban streams (Greeves, 2023).

I hypothesized that the altered environmental conditions associated with Urban Stream Syndrome reduces the presence of intolerant macroinvertebrate taxa, ultimately altering environmental processes via a reduction in functional feeding diversity. My results supported my hypothesis, as I found that Gross Creek was physically distinguishable from Hudson's Branch consistent with being impacted by the Urban Stream Syndrome and lacked any macroinvertebrates with a tolerance value under 5. In contrast Hudson's Branch, the non-urban stream, contained intolerant macroinvertebrates with tolerance values ranging from 0 to 8. Other studies have also found a reduction in intolerant macroinvertebrates in urban streams (Anderson & Day, 1986, Chessman & McEvoy, 2012, Barnum et al., 2017). Despite these differences in the

presence of intolerant taxa in Hudson's Branch, chironomids still dominated the macroinvertebrate communities of both streams. This finding is consistent with the results found by Violin et al., (2011), who also found that Chironomidae dominated both urbanized and restored streams in their study. Chironomids have a tolerance level of 8, the highest tolerance level collected in this study. Because of their high tolerance, I weighted the mean tolerances of each sample by the abundance of each taxon. Since the chironomids were the most abundant taxa collected from both streams, the abundance-weighted means were more similar in the streams but still distinguished the creeks and indicated a reduction in intolerant taxa in Gross Creek.

I found that tolerant and in-tolerant macroinvertebrates were largely equally distributed between the sampled stream habitats, and the observed tolerance levels largely followed the patterns of the site overall. This observation is consistent with results found by Chessman and McEvoy (2011), who found that there were not significant differences in macroinvertebrate tolerance among habitats as a result of the tested stressors. In all habitats in both creeks, the maximum tolerance observed was always 8, because chironomids were present in all samples. In Hudson's Branch, the minimum tolerance was 0 for pools and 1 for riffles and runs, but these values were contributed by multiple unique taxa, meaning multiple taxa with low level tolerances were found in all three habitats in the nonurban stream. In contrast, the least tolerant organism found in all habitats in Gross Creek was always the larvae in the genus *Anchytarsus*, meaning only one taxon with a medium tolerance was found in the urban stream. This reduction of diversity in tolerance and taxonomic composition in Gross Creek can be associated with the effects of the Urban Stream Syndrome, as one of the expected outcomes is reduced intolerant species and therefore, reduced taxonomic diversity (Chessman & McEvoy, 2012).

When using the abundance-weighted tolerance in the habitats, I once again found that, the dominance of the chironomids makes the difference between the mean tolerance of Hudson's Branch and Gross Creek smaller. Notably, both of the streams were dominated by the most tolerant species, even though Hudson's Branch is considered "un-impacted" for the purposes of this study. However, Hudson's Branch, though non-urban when compared to Gross Creek, is still recovering from nearby agricultural land uses characteristic of central Virginia, which can alter macroinvertebrate community structure, and is likely impacted by the developed land use, given its proximity to Hampden-Sydney College (Sterling et al., 2016, Stoughton, 2017).

My results support my hypothesis that the changes in environmental conditions that result from the Urban Stream Syndrome would eliminate intolerant groups of macroinvertebrates. My findings also supported my hypothesis that the altered environmental conditions due to the impacts of Urban Stream Syndrome would reduce taxonomic diversity, and functional diversity in the urban stream. Total macroinvertebrate density did not differ significantly between the creeks but this is unsurprising, as Chironomidae dominated both systems and were the most abundant taxa. Taxa richness, however, did differ between sites. The number of unique taxa in each site was impacted by urbanization, through the simplification of taxonomic composition in Gross Creek, the urban stream (Sterling et al., 2016). Interpretation of diversity difference between the creeks must be done with some caution, since there is some unknown quantity of diversity that remains unquantified as I was unable to identify all taxa down to species, with many identified down to genus and specifically the chironomids which I was only able to identify down to subfamily.

For this study, both Inverse Simpson diversity and Shannon diversity indices were used as they are commonly used indices in the field of ecology (Hill, 1973) and to ensure that my

conclusions were not influenced by choice of index. Hudson's Branch had greater taxonomic diversity than Gross Creek when using Inverse Simpson Taxonomic diversity and the Shannon Taxonomic diversity indices. This difference further indicates that the Urban stream syndrome impacts the stream through the simplification of taxonomic diversity (Sterling et al., 2016). The Taxonomic diversity also differed amongst the habitats when using the Inverse Simpson and Shannon indices, where riffle habitats had the highest taxonomic diversity and pools had the lowest taxonomic diversity.

My results were consistent with my hypothesis that the changes in environmental conditions that result from the Urban Stream Syndrome would ultimately reduce functional diversity through a reduction of taxonomic diversity. I found that the functional diversity was significantly greater in the non-urban stream. All feeding groups were represented in Hudson's Branch. In contrast, Gross Creek, the urban stream, with a significantly lower representation of Shredders, Filterers, and Scrapers in Gross Creek (Peng et al., 2024).

The NMDS Ordination final stress results indicate that the ordinations can be interpreted without the risk of inaccurate conclusions being drawn (Clarke, 1993). The NMDS ordination based on macroinvertebrate taxa showed that macroinvertebrate community composition differed between the urban and non-urban streams and also among habitats within the stream. The macroinvertebrate samples from the urban and non-urban streams separate on the first axis of the NMDS ordination (NMDS1), indicating differences in the macroinvertebrate communities but there was overlap between the streams in samples taken from pool habitats. This overlap could be explained by the significantly dominant presence of chironomids in pools in both Hudson's Branch and Gross Creek. These findings are comparable to the findings of Ferrington et al., (1995), who found that pool habitats were dominated by chironominae. Pool habitats can be

characterized by their low flow and low dissolved oxygen content, both of which can be challenging conditions for some taxa, leading the highly tolerant chironomids to dominate the habitat (Ferrington et al., 1995). Along with the other taxonomic diversity data described above, the ordination results show that the macroinvertebrate community of Hudson's Branch, the non-urban stream is different from what is found Gross Creek, the urban stream.

Similarly, the NMDS ordination based on functional feeding groups showed that the functional communities of the streams differed. The urban and non-urban streams separate on the first axis if the NMDS ordination (NMDS1), supporting a difference functional composition. Unlike the ordination based on taxa, the separation was evident in all habitats, although the least amount of separation was still found in pool habitats. Again, these ordination results support the conclusion that Hudson's Branch and Gross Creek's functional communities are different (Gao et al., 2023.

I hypothesized that the lack of intolerant species present in urban systems would result in a decrease in the decomposition of organic matter due to the reduction in functional diversity (Tampo et al., 2021). I estimated organic matter processing by macroinvertebrates using an estimate of the organic matter consumption rate from Stagliano and Whiles (2002). I focus my analysis organic matter consumption because macroinvertebrate feeding represents a major pathway of organic matter processing in streams (citation needed). The organic matter consumption rates from Stagliano and Whiles (2002) come from a prairie stream, which may not fully represent the macroinvertebrate consumption rates of the piedmont streams in my study. Ross and Wallace (1983) found that the consumption rate for filtering Hydropsychid caddis flies ranged between 2.4 and 11.7 g AFDM m-2 y-1 in a forested mountain stream, which brackets the mean filterer consumption rate of 2.9 g AFDM m-2 y-1 reported by Stagliano and Whiles

(2002). Although they did not analyze consumption by functional group, Rosi-Marshall and Wallace (2002) found that consumption rates within the food webs of mountain rivers were primarily between 0 and 20 g AFDM $m^{-2}$ $y^{-1}$. The similarity of the consumption rates reported across this diversity of systems, suggest that our estimates are informative if interpreted with appropriate caution.

According to the model, organic matter processing of the sites was highly variable. The majority of organic matter processing in Gross Creek was due to Collectors. Chironomids, being the most tolerant taxa found in this study, have dominated the urban stream and naturally are responsible for the highest amount of organic matter processing as Collectors. In Hudson's Branch, however, the greatest mean organic matter processing was due to Shredders. This shift in group responsible for the most organic matter processing is interesting because although chironomids were also the dominant taxa found in the non-urban stream, they were not responsible for the highest amount of organic matter processing as they were in the urban stream. This shift is highlighted by the fact that the Shredders were responsible for 5.4 times more consumption than the Collectors in Hudson's Branch. While Gross Creek had more organic matter processing than Hudson's Branch when focusing on Collectors and Predators, with Collectors being 1.8 times greater and Predators being 1.7 times greater, the most significant differences between the sites were in Shredding, where the consumption was 28 times greater in Hudson's Branch than in Gross Creek, and Filtering, where the consumption was 15 times greater in Hudson's Branch than Gross Creek. This difference in consumption between the sites suggests that the urban stream has a significant reduction in organic matter processing pathways when compared to the non-urban stream. This change in consumption rates tells a story of a shift from a shredder dominated non-urban system to a collector dominated system when impacted by

urbanization. Significantly, Shredders are responsible for the breakdown of detritus (Graça et al., 2001). In Southeastern streams, shredding is a dominant form of organic matter processing, making the reduced presence of Shredders in Gross Creek a significant loss of organic matter breakdown (Oester et al., 2021). The low representation of Shredders in the urban stream supports the conclusion that Gross Creek's taxonomic diversity and functional diversity is simplified when compared to Hudson's Branch. In a stream impacted by Urban Stream Syndrome where the dominant function has shifted from shredding to collecting, the system suffers from a major reduction in the breakdown of detritus.

In Hudson's Branch, the mean organic matter processing by Collectors and Predators was uniform across habitats, but in Gross Creek Collectors and Predators were responsible for the most organic matter consumption of the pools and the runs. When first developing my hypothesis, I hypothesized that Predators would have a reduced presence in the non-urban stream. However, I found that the Predators were primarily tanypodinae, a tolerant chironominae, they maintained the function in the stream. Notably, their prey is primarily meiofauna, so although they are fulfilling the function of predation, there are still larger organisms that are not being predated upon (Schmid & Schmid-Araya, 1997). The pool habitat is mostly comprised of sediment and has a lower flow than other habitats, making the pool less suitable for some taxa, and ultimately resulting in decreased diversity (Anderson & Day, 1986). Given these habitat characteristics, the highly tolerant chironomids making up the Collector and Predator feeding groups have no trouble dominating this habitat.

In Hudson's Branch the mean organic matter processing by Filterers, Shredders, and Scrapers was greatest in the Riffles. The Filterers, Shredders, and Scrapers were mostly made up of taxa with lower tolerance values, and as such had a reduced presence in Gross Creek. In Gross

Creek, mean organic matter processing by Filterers was almost nonexistent, with a small presence in Riffle habitats and no presence in Pools and Runs. Similarly, scrapers were completely unrepresented in all habitats of the system. The small amount of consumption by Filterers and complete lack of consumption from Scrapers means there was a significant reduction of organic matter processing pathways in the system.

## Conclusion

All of these results suggest that the Urban stream syndrome impacts streams through altered organic matter processing due to the reduction of functional and taxonomic diversity. These results support my hypothesis that the altered environmental conditions associated with Urban Stream Syndrome will reduce the presence of intolerant macroinvertebrate taxa, ultimately altering environmental processes via reduction in functional feeding diversity. I found that there was a reduction in taxonomic diversity in Gross Creek when compared to Hudson's Branch, which included a reduction in the presence of intolerant species. I also found that the reduction in intolerant species resulted in a reduced presence of certain feeding groups in the urban stream, primarily the Filterers, Shredders, and Scrapers. As a result of the limited presence of those feeding groups, there was an overall reduction of consumption by Filterers, Shredders, and Scrapers in Gross Creek, ultimately resulting in a reduction of organic matter processing pathways in the urban stream.

## Figures

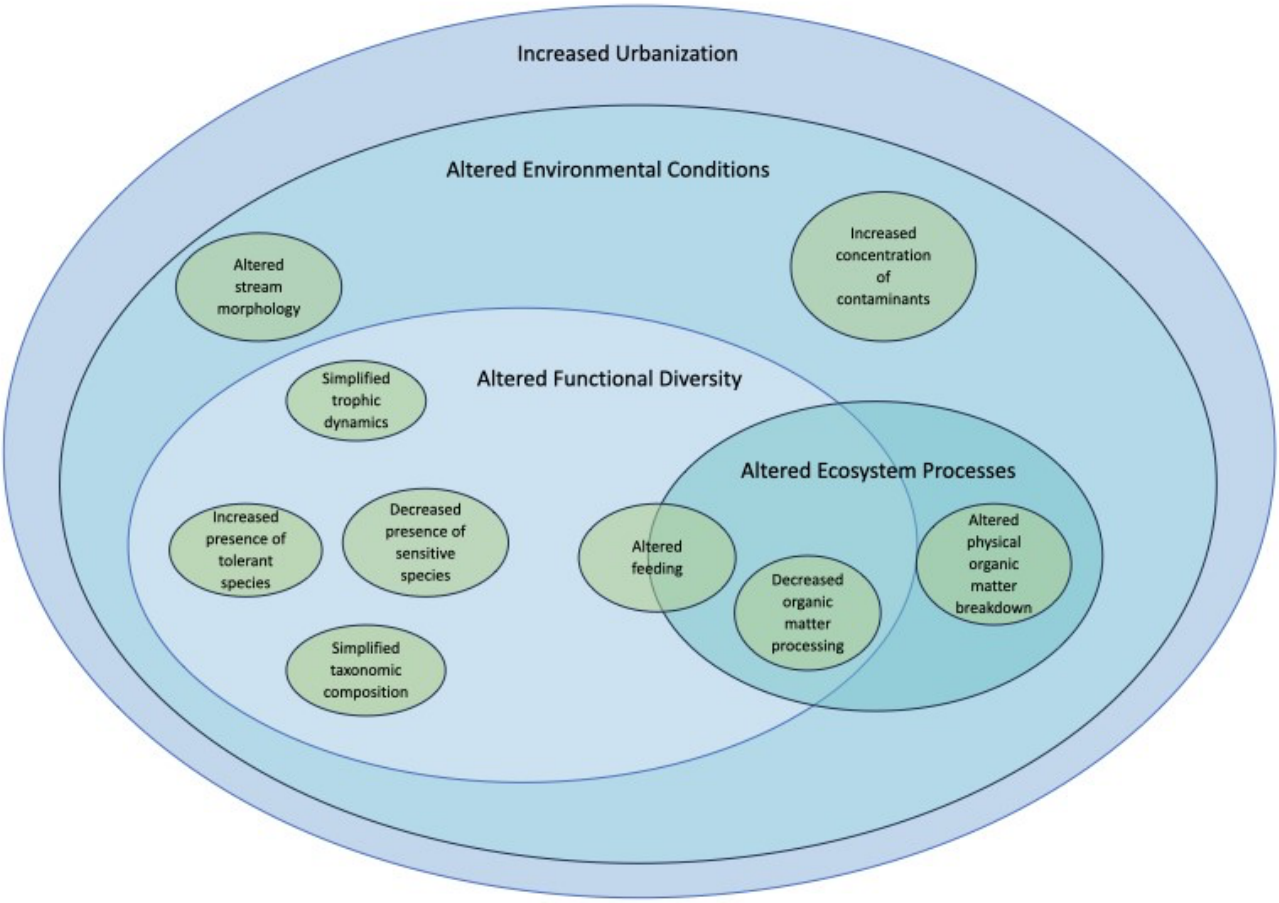


Figure 1. A conceptual illustration of the hypothesized impact of Urban Stream Syndrome on stream ecosystem processes through altered environmental conditions and functional diversity. Increased urbanization (the large blue circle) results in changes in environmental conditions (the lighter blue circle), which then affect ecosystem processes either directly, or through modifications to functional diversity (the lightest blue circle). The specific mechanisms of action are shown by the green circles.

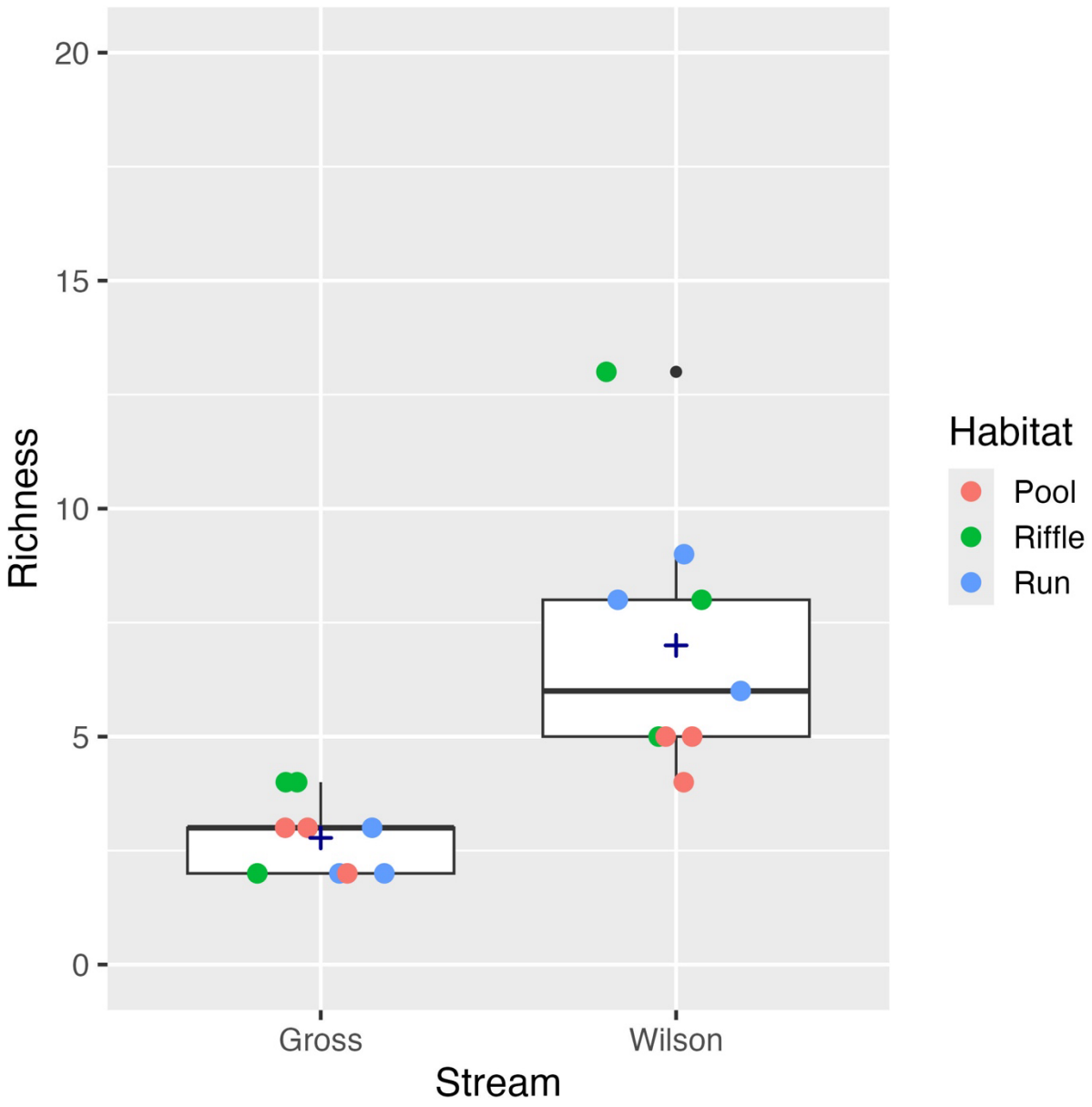


Figure 2. Taxonomic richness in the riffle, run, and pool habitats of Gross Creek and Hudson's Branch. The boxplots show the median as the bold line with the box representing the 1st and 3rd quartiles. The whiskers indicate the minimum and maximum values to within 1.5-times the interquartile range. Values outside of that are shown as outlies with black dots. Each dot represents the value of a single sample where riffle habitats are represented by green dots, run habitats are represented by blue dots, and pool habitats are represented by coral dots. The mean values are represented by the dark blue cross sign.

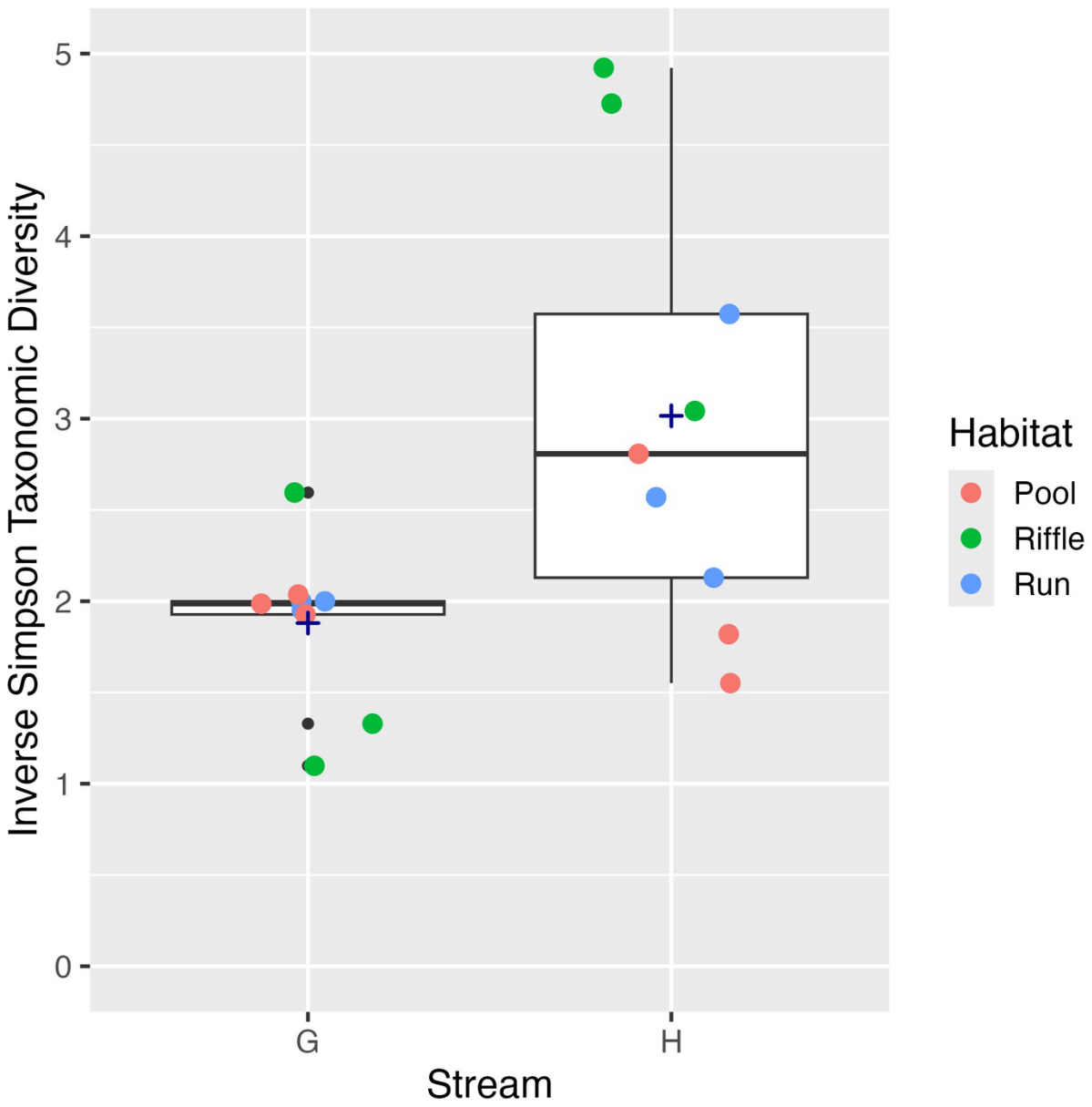


Figure 3. Inverse Simpson taxonomic diversity in the riffle, run, and pool habitats of Gross Creek and Hudson's Branch. The boxplots show the median as the bold line with the box representing the 1st and 3rd quartiles. The whiskers indicate the minimum and maximum values to within 1.5-times the interquartile range. Values outside of that are shown as outlies with black dots. Each dot represents the value of a single sample where riffle habitats are represented by green dots, run habitats are represented by blue dots, and pool habitats are represented by coral dots. The mean values are represented by the dark blue cross sign.

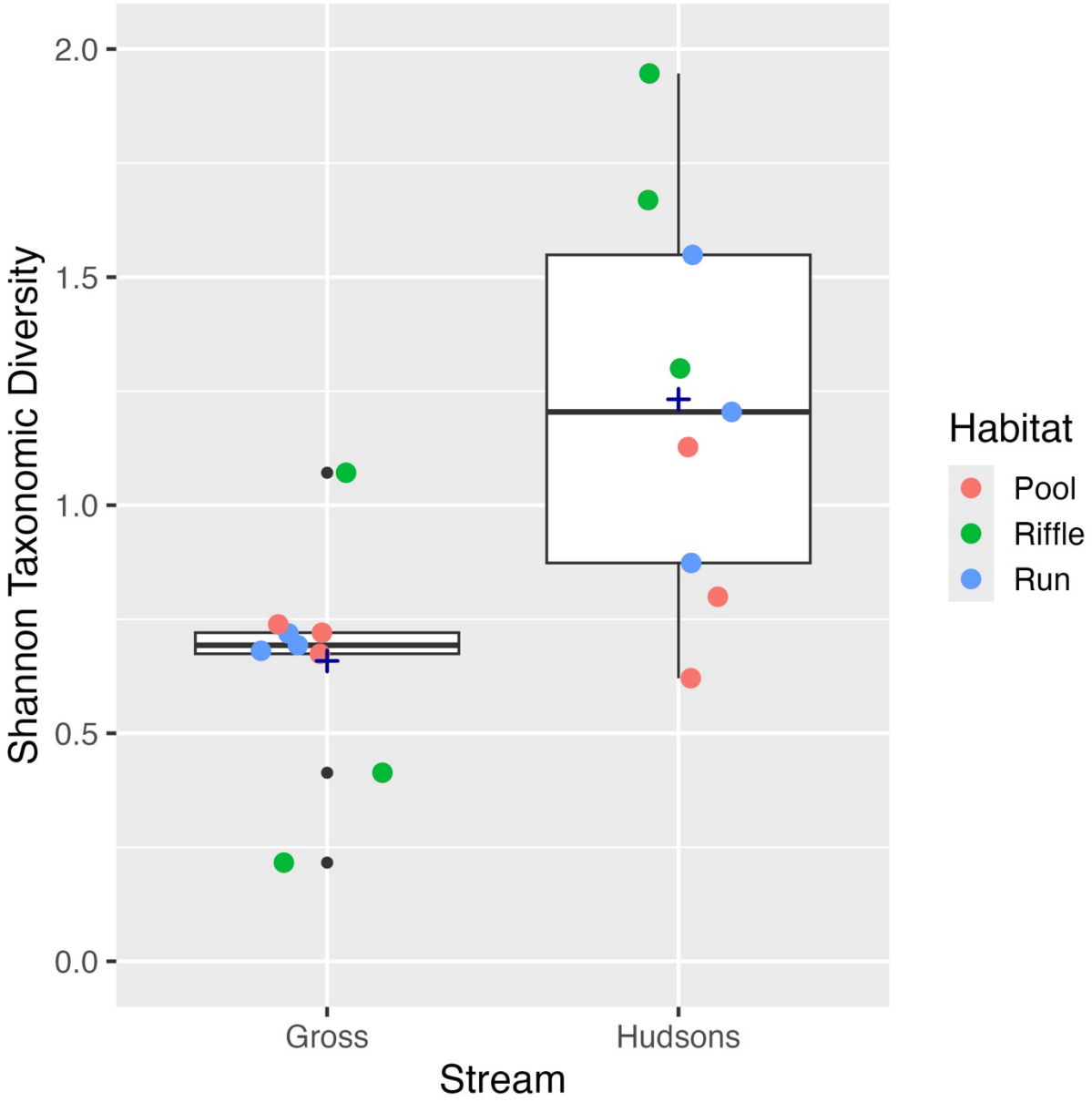


Figure 4. Shannon taxonomic diversity in the riffle, run, and pool habitats of Gross Creek and Hudson's Branch. The boxplots show the median as the bold line with the box representing the 1st and 3rd quartiles. The whiskers indicate the minimum and maximum values to within 1.5-times the interquartile range. Values outside of that are shown as outlies with black dots. Each dot represents the value of a single sample where riffle habitats are represented by green dots, run habitats are represented by blue dots, and pool habitats are represented by coral dots. The mean values are represented by the dark blue cross sign.

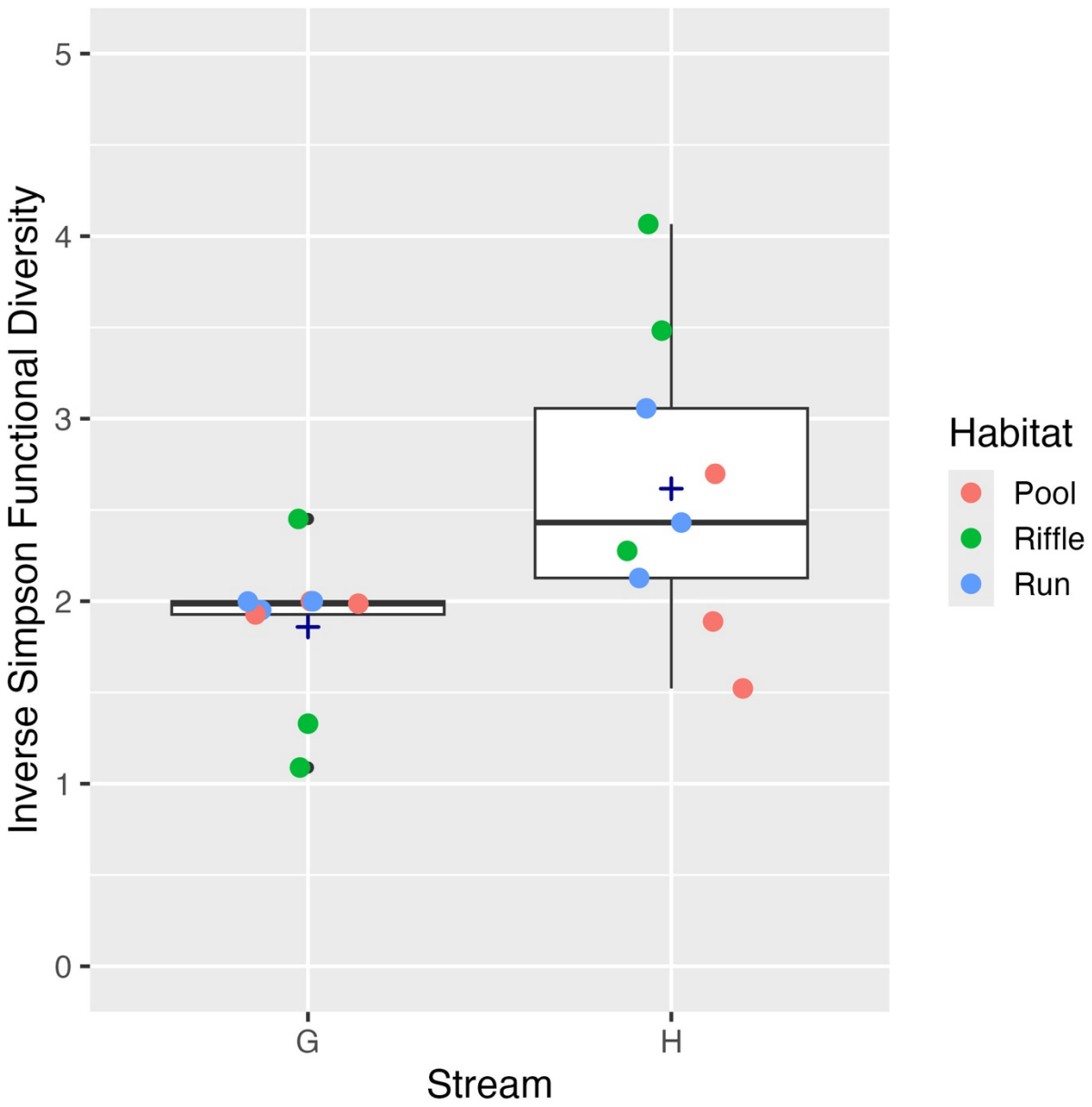


Figure 5. Inverse Simpson functional diversity in the riffle, run, and pool habitats of Gross Creek and Hudson's Branch. The boxplots show the median as the bold line with the box representing the 1st and 3rd quartiles. The whiskers indicate the minimum and maximum values to within 1.5-times the interquartile range. Values outside of that are shown as outlies with black dots. Each dot represents the value of a single sample where riffle habitats are represented by green dots, run habitats are represented by blue dots, and pool habitats are represented by coral dots. The mean values are represented by the dark blue cross sign.

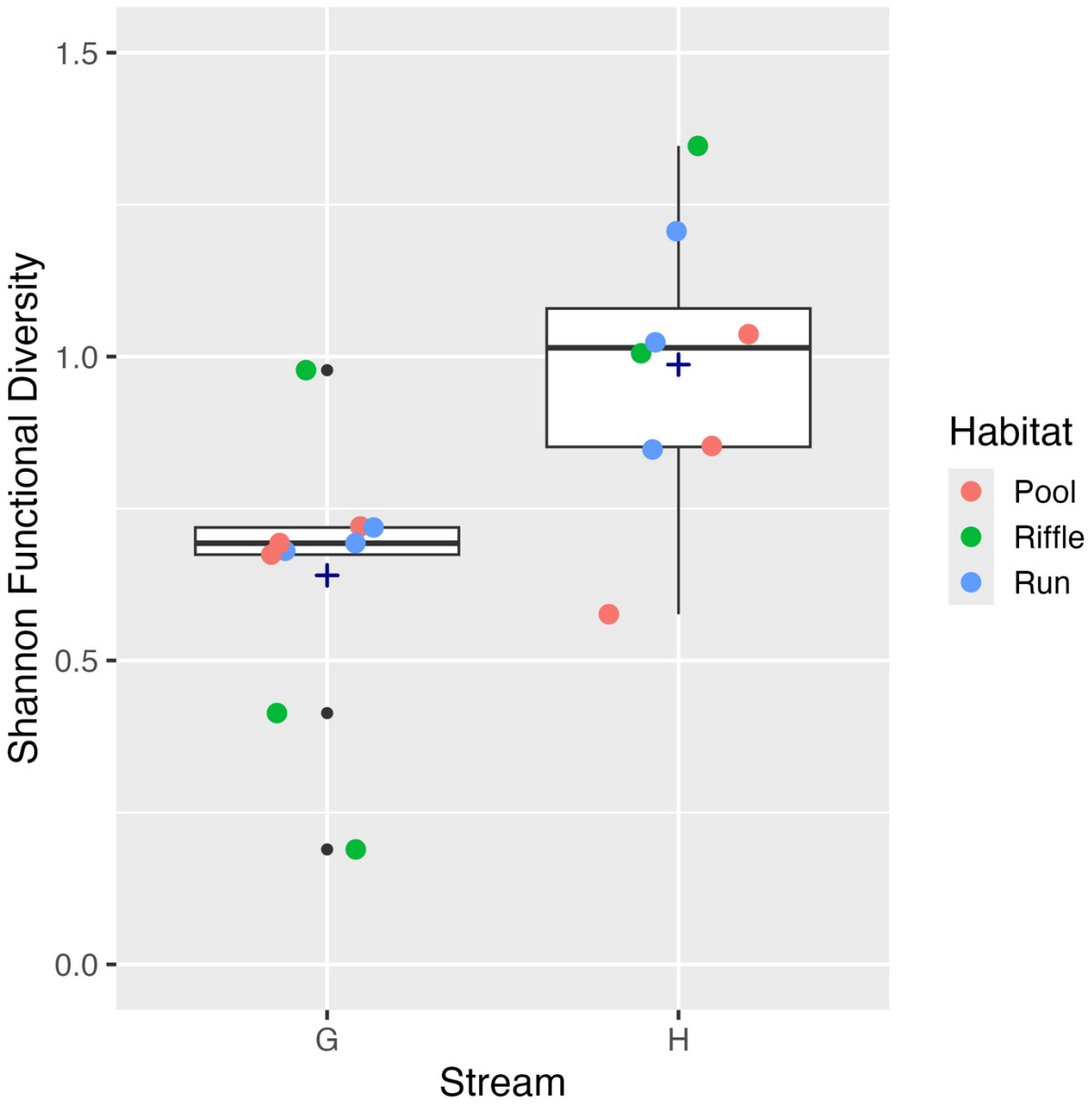


Figure 6. Shannon functional diversity in the riffle, run, and pool habitats of Gross Creek and Hudson's Branch. The boxplots show the median as the bold line with the box representing the 1st and 3rd quartiles. The whiskers indicate the minimum and maximum values to within 1.5-times the interquartile range. Values outside of that are shown as outlies with black dots. Each dot represents the value of a single sample where riffle habitats are represented by green dots, run habitats are represented by blue dots, and pool habitats are represented by coral dots. The mean values are represented by the dark blue cross sign.

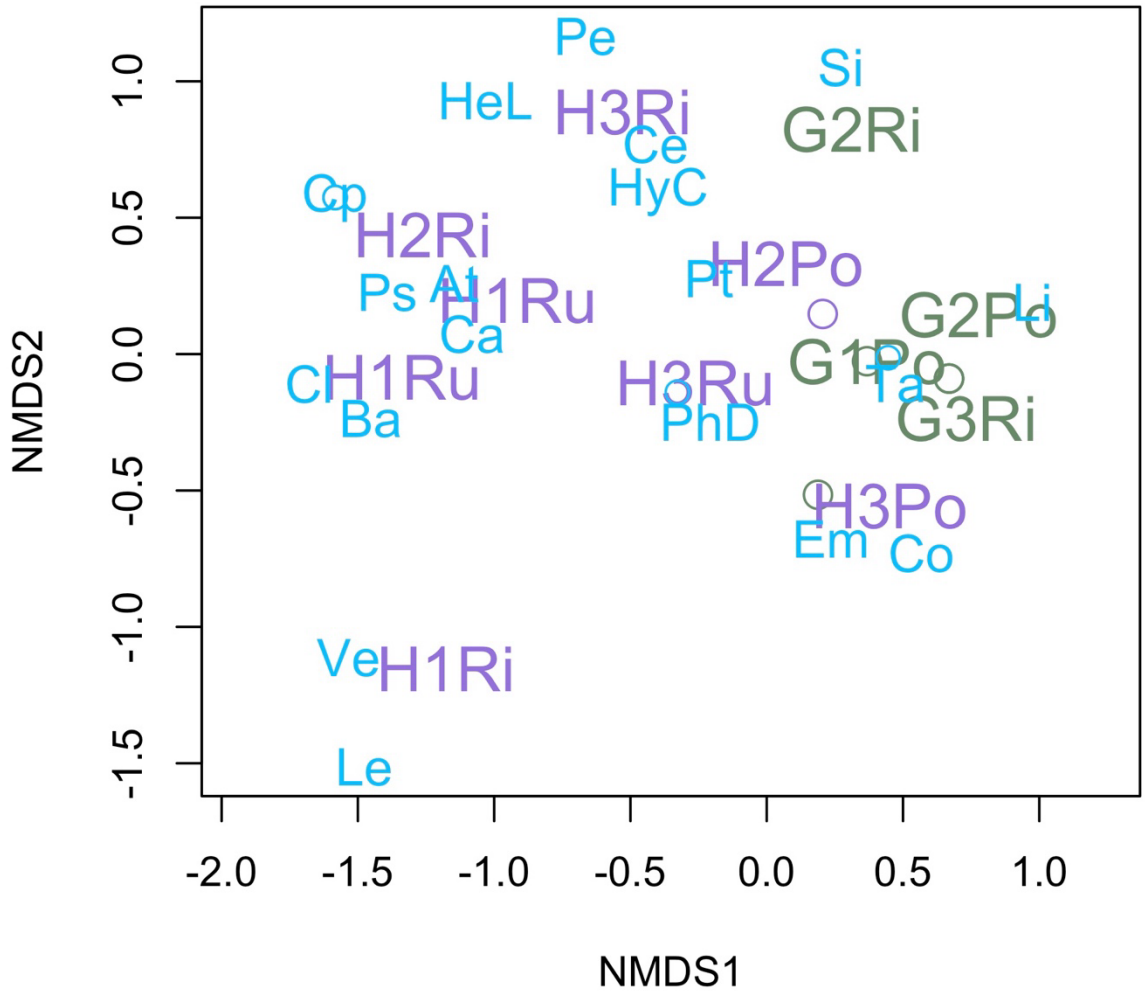


Figure 7. NMDS ordination based on taxa showing the loading of each sample on the first (NMDS1) and second (NMDS2) axes of the ordination. The samples are coded by stream (H = Hudson's, G = Gross), habitat (Ri = riffle, Ru = run, and Po = pool), and replicate number. Where marker overlap prevented the printing of the sample identity, the location of the sample is shown as an open circle. The Gross Creek samples are in green, the Hudson's Branch samples are plotted in purple. The loadings of the abbreviated names of taxa used in the ordination are shown in blue. The taxa abbreviations are Appendix 1.

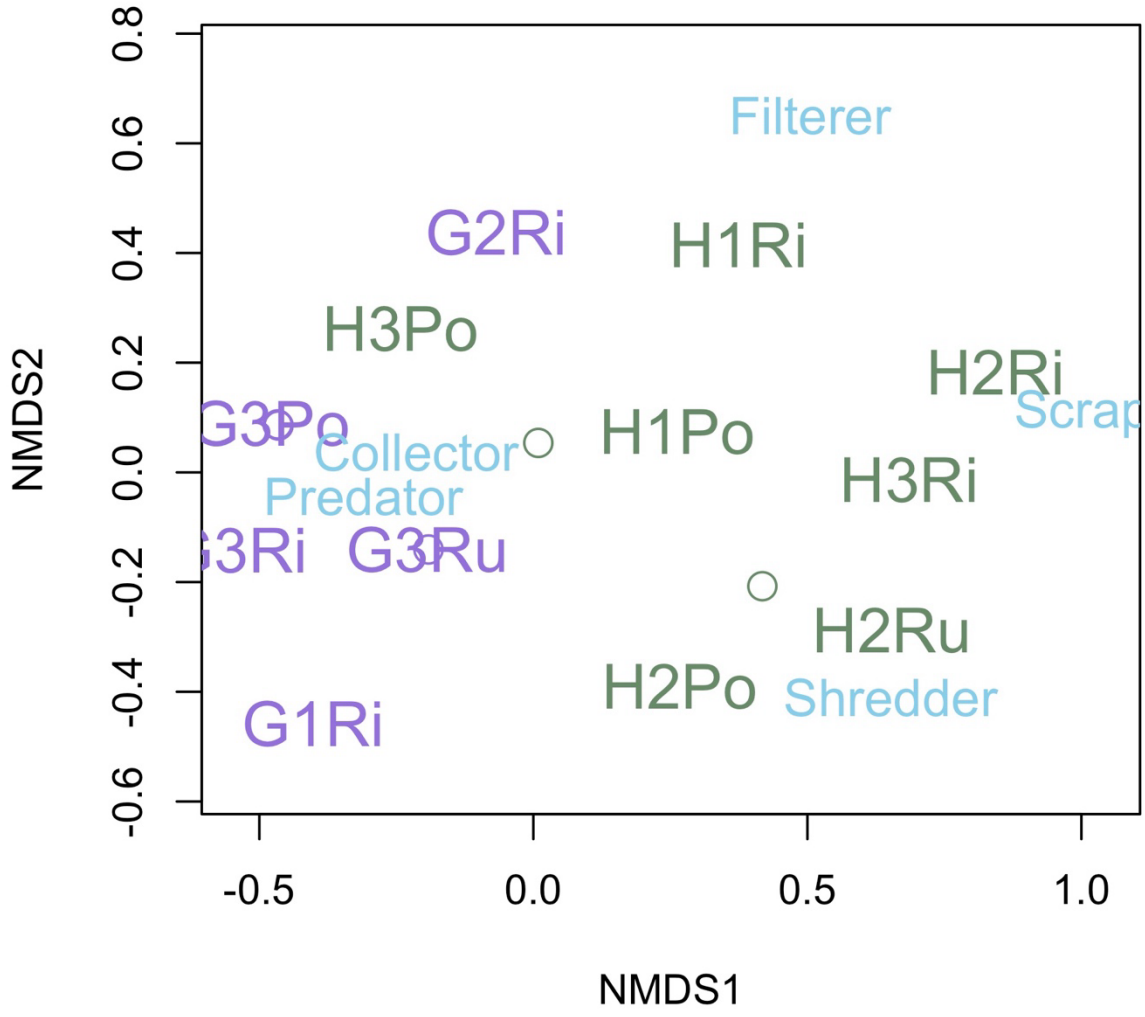


Figure 8. NMDS ordination based on function showing the loading of each sample on the first (NMDS1) and second (NMDS2) axes of the ordination. The samples are coded by stream (H = Hudson's, G = Gross), habitat (Ri = riffle, Ru = run, and Po = pool), and replicate number. Where marker overlap prevented the printing of the sample identity, the location of the sample is shown as an open circle. The Gross Creek samples are plotted in green, the Hudson's Branch samples are plotted in purple. The loadings of the functional feeding groups used in the ordination are shown in blue.

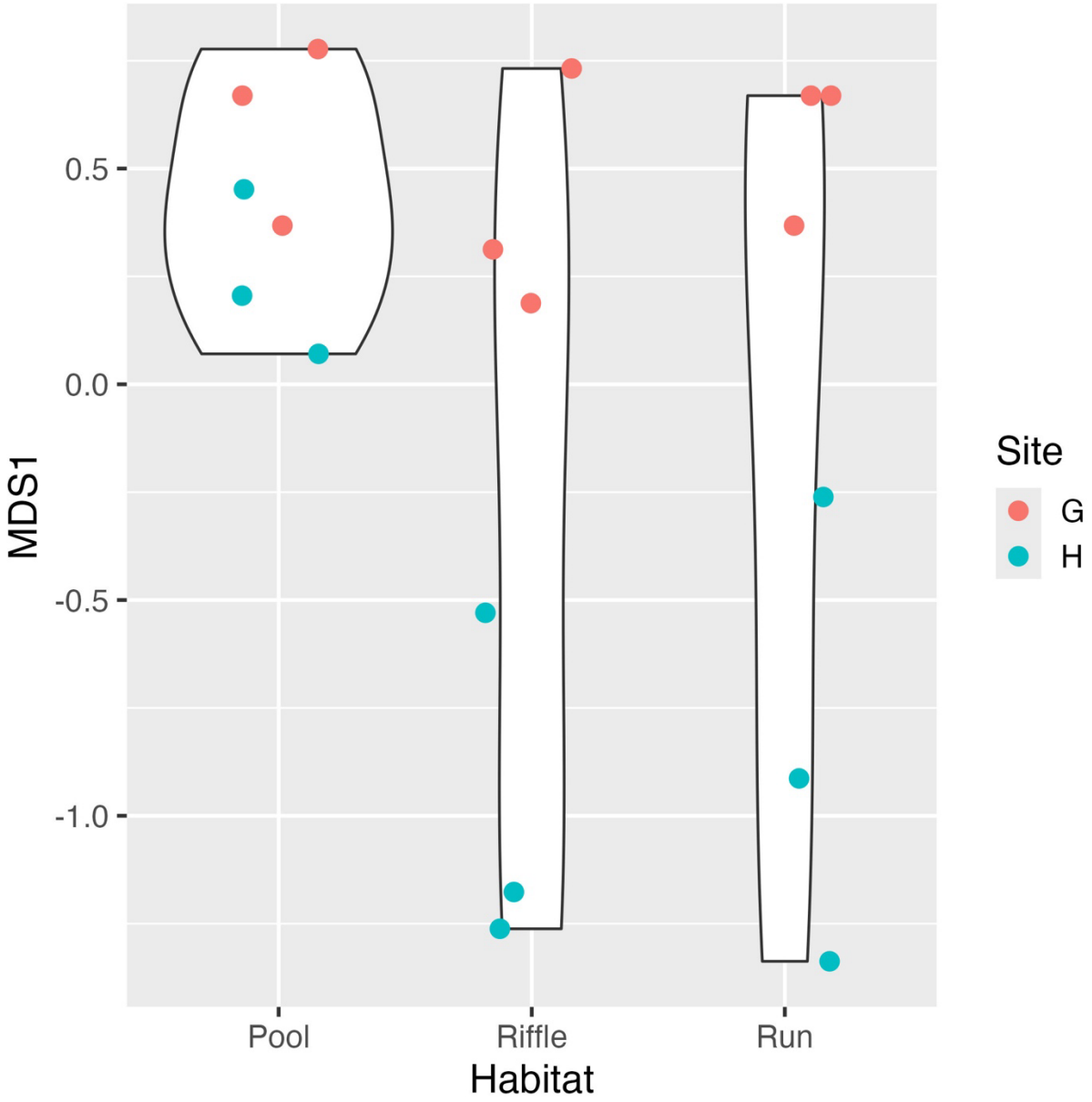


Figure 9. Violin plot of the first axis of variation for the NMDS ordination based on taxa for riffle, run, and pool habitats in Gross Creek and Hudson's Branch. The width of the "violin" shows the probability density of the data and individual samples are shown as points, where Gross Creek is represented by the coral dots and Hudson's Branch is represented by the teal dots.

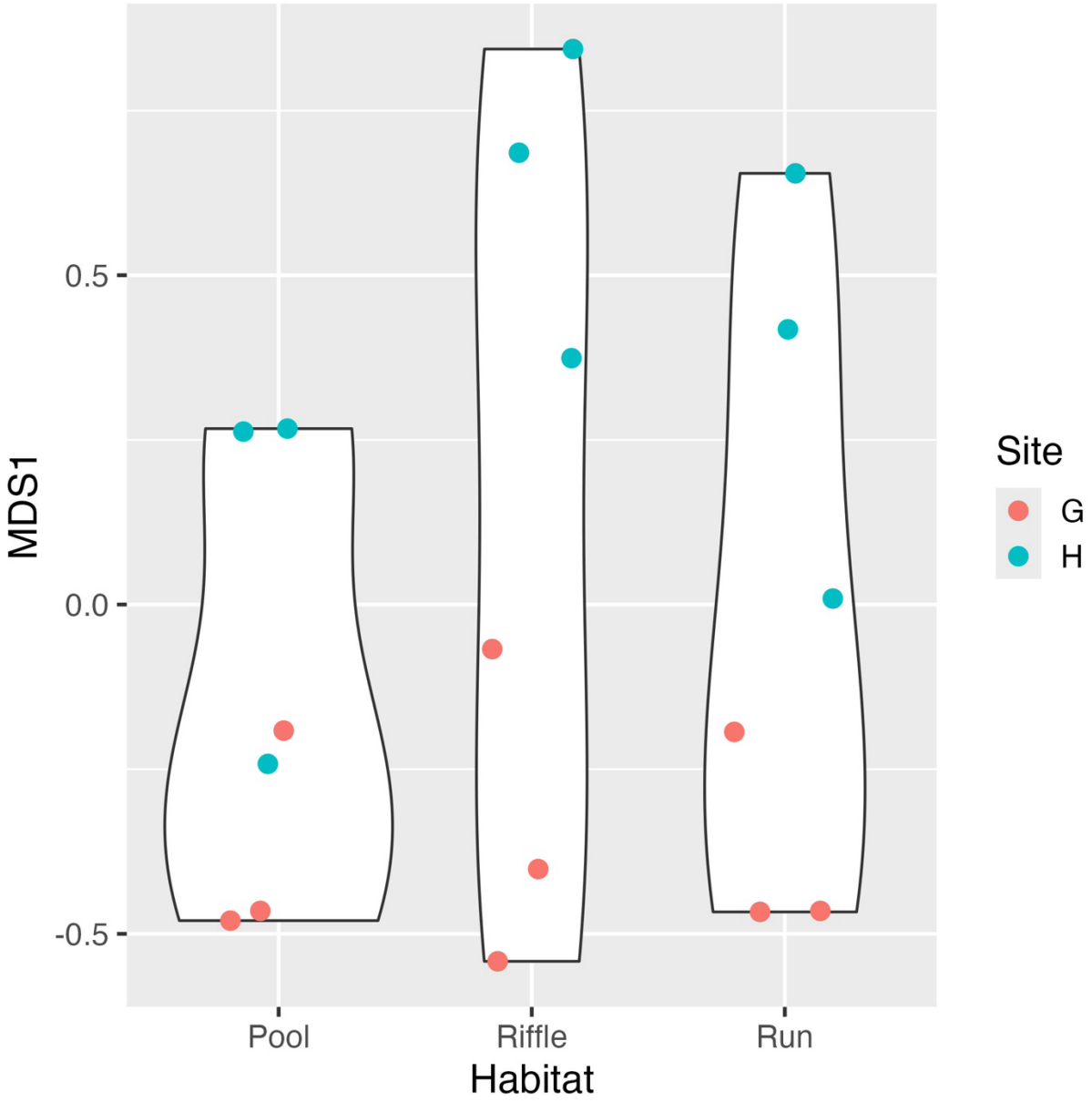


Figure 10. Violin plot of the first axis of variation for the NMDS ordination based on functional feeding groups for riffle, run, and pool habitats in Gross Creek and Hudson's Branch. The width of the "violin" shows the probability density of the data and individual samples are shown as points, where Gross Creek is represented by the coral dots and Hudson's Branch is represented by the teal dots.

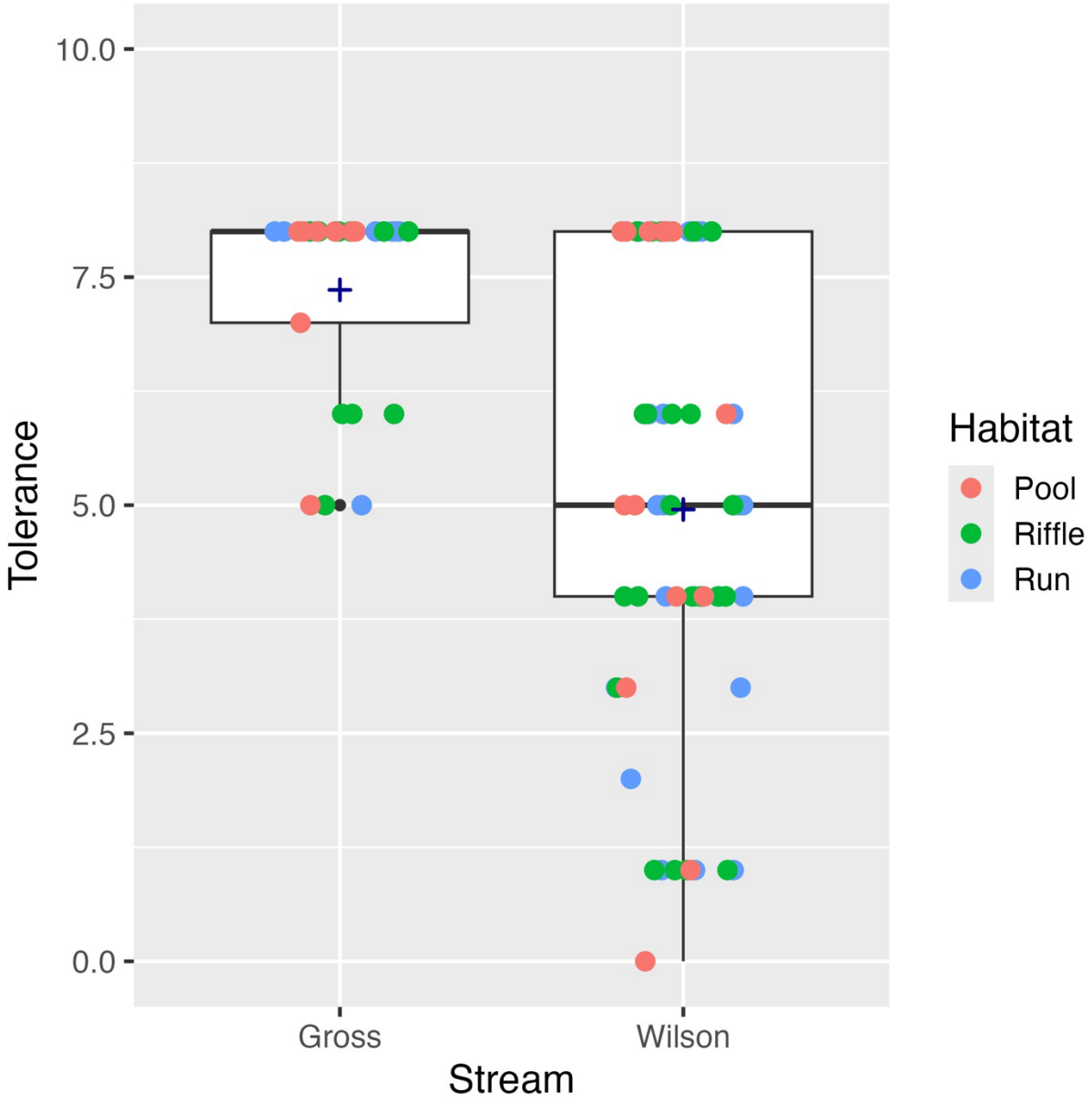


Figure 11. Un-weighted tolerance levels in the riffle, run, and pool habitats of Gross Creek and Hudson's Branch. The boxplots show the median as the bold line with the box representing the 1st and 3rd quartiles. The whiskers indicate the minimum and maximum values to within 1.5-times the interquartile range. Values outside of that are shown as outlies with black dots. Each dot represents the value of a single sample where riffle habitats are represented by green dots, run habitats are represented by blue dots, and pool habitats are represented by coral dots.

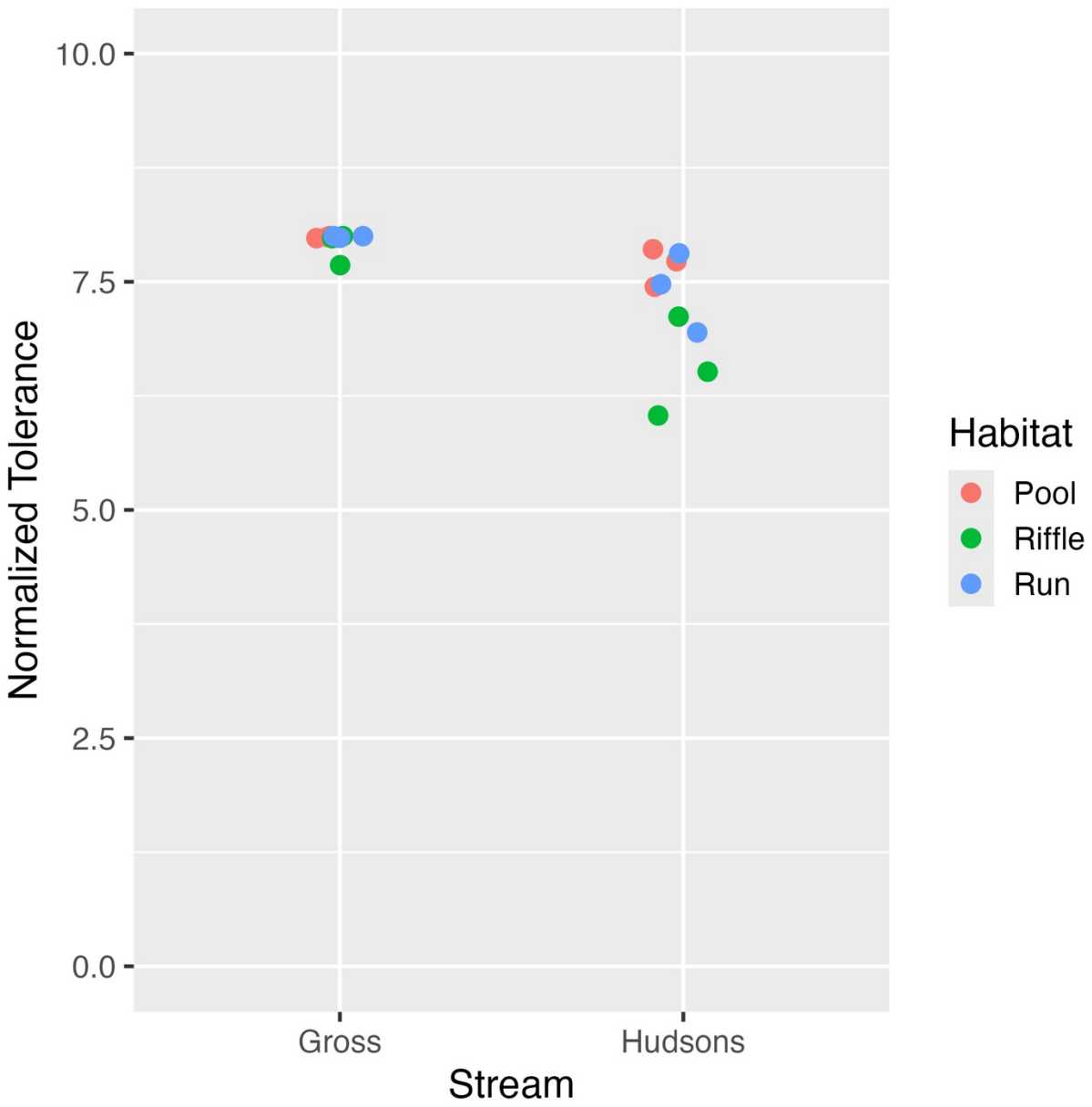


Figure 12. Abundance-weighted mean tolerance values for riffle, run, and pool habitats of Gross Creek and Hudson's Branch. Each dot represents the value of a single sample where riffle habitats are represented by green dots, run habitats are represented by blue dots, and pool habitats are represented by coral dots.

## Tables

Table 1. The mean abundance (number of individuals), standard deviation of mean abundance, and the samples used to calculate the mean (N) found in each habitat of each site. Taxa are identified by order, family, and genus. Taxa that were not identified to genus are indicated with "NA". Note that taxa in the Chironomidae family are identified as belonging to the subfamily Chironominae or Tanypodinae. Tolerance indicates the tolerance level for that taxon reported by Purdue (Speelman & Carroll, 2016).

| *Site* | *Habitat* | *Order* | *Family* | *Genus* | *Abundance mean* | *Abundance standard dev* | *N* | *Tolerance* |
|---|---|---|---|---|---|---|---|---|
| Hudson | Run | Diptera | Chironomidae (Chironominae) | NA | 62.67 | 35.91 | 3 | 8 |
| | | Diptera | Chironomidae (Tanypodinae) | NA | 62 | 49.27 | 3 | 8 |
| | | Diptera | Ceratopogonidae | NA | 2.5 | 0.71 | 2 | 6 |
| | | Diptera | Athericidae | Atherix | 2 | | 1 | 2 |
| | | Coleoptera | Ptilodactylidae | Anchytarsus | 5.33 | 3.06 | 3 | 5 |
| | | Coleoptera | Psephenidae | Psephenus | 2.5 | 2.12 | 2 | 4 |
| | | Ephemeroptera | Baedidae | Baetis | 3 | 1.41 | 2 | 4 |
| | | Plecoptera | Capniidae | Allocapnia | 2 | 0 | 3 | 1 |
| | | Hemiptera | Veliidae | Rhagovelia | 1 | | 1 | 6 |
| | | Odonata | Calopterygidae | Hetaerina | 1 | | 1 | 5 |
| | | Trichoptera | Philopotamidae | Dolophilodes | 2 | | 1 | 3 |
| | | Trichoptera | Philopotamidae | Chimarra | 2 | | 1 | 3 |
| | Riffle | Diptera | Chironomidae (Chironominae) | NA | 48.33 | 45.65 | 3 | 8 |
| | | Diptera | Chironomidae (Tanypodinae) | NA | 32.67 | 21.94 | 3 | 8 |
| | | Trichoptera | Lepidostomidae | Lepidostoma | 1 | | 1 | 1 |
| | | Ephemeroptera | Baedidae | Baetis | 7 | 8.49 | 2 | 4 |
| | | Hemiptera | Veliidae | Rhagovelia | 2 | | 1 | 6 |
| | | Diptera | Simuliidae | Simulium | 1 | 0 | 2 | 6 |
| | | Trichoptera | Philopotamidae | Dolophilodes | 16 | | 1 | 3 |
| | | Trichoptera | Hydropsychidae | Cheumatopsyche | 29.5 | 13.44 | 2 | 4 |
| | | Trichoptera | Hydropsychidae | Arctopsyche | 4 | | 1 | 4 |
| | | Coleoptera | Ptilodactylidae | Anchytarsus | 28 | 4.24 | 2 | 5 |
| | | Coleoptera | Psephenidae | Psephenus | 4 | | 1 | 4 |
| | | Ephemeroptera | Heptageniidae | Leucrocuta | 3 | 1.41 | 2 | 4 |
| | | Ephemeroptera | Heptageniidae | Stenacron | 6 | | 1 | 4 |
| | | Plecoptera | Capniidae | Allocapnia | 4 | | 1 | 1 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Plecoptera | Chloroperidae | Alloperla | 6 | | 1 | 1 |
| | | Diptera | Ceratopogonidae | NA | 19 | | 1 | 6 |
| | | Plecoptera | Perlidae | Paragnetina | 5 | | 1 | 1 |
| | Pool | Diptera | Chironomidae (Chironominae) | NA | 43 | 7.55 | 3 | 8 |
| | | Diptera | Chironomidae (Tanypodinae) | NA | 82 | 99.58 | 3 | 8 |
| | | Trichoptera | Hydrosycidae | cheumatopsyche | 2 | 1.41 | 2 | 4 |
| | | Coleoptera | Ptilodactylidae | Anchytarsus | 10 | 11.31 | 2 | 5 |
| | | Diptera | Ceratopogonidae | NA | 1 | | 1 | 6 |
| | | Trichoptera | Philopotamidae | Dolophilodes | 4 | | 1 | 3 |
| | | Plecoptera | Capniidae | Allocapnia | 6 | | 1 | 1 |
| | | Megaloptera | Corydalidae | Nigronia | 2 | | 1 | 0 |
| Gross | Run | Diptera | Chironominae | NA | 143.33 | 106.29 | 3 | 8 |
| | | Diptera | Tanypodinae | NA | 125.67 | 99.75 | 3 | 8 |
| | | Coleoptera | Ptilodactylidae | Anchytarsus | 2 | | 1 | 5 |
| | Riffle | Diptera | Chironomidae (Chironominae) | NA | 11.67 | 3.06 | 3 | 8 |
| | | Diptera | Chironomidae (Tanypodinae) | NA | 104 | 106.97 | 3 | 8 |
| | | Diptera | Empididae | Hemerodromia | 1 | | 1 | 6 |
| | | Coleoptera | Ptilodactylidae | Anchytarsus | 1 | | 1 | 5 |
| | | Diptera | Ceratopogonidae | NA | 1 | | 1 | 6 |
| | | Diptera | Simuliidae | Simulium | 6 | | 1 | 6 |
| | Pool | Diptera | Chironomidae (Chironominae) | NA | 116 | 41.33 | 3 | 8 |
| | | Diptera | Chironomidae (Tanypodinae) | NA | 91.33 | 28.87 | 3 | 8 |
| | | Coleoptera | Ptilodactylidae | Anchytarsus | 1 | | 1 | 5 |
| | | Odonata | Libellulidae | Libellula | 2 | | 1 | 7 |

Table 2. The mean (+/- 1 SD) Inverse Simpson Taxonomic Diversity and mean Shannon Taxonomic Diversity of each stream and of each habitat type.

| | *Inverse Simpson* | *Shannon* |
|---|---|---|
| Gross Cr | 1.88 (+/- 0.433) | 0.659 (+/- 0.235) |
| Hudson's Br. | 3.02 (+/- 1.20) | 1.23 (+/- 0.434) |
| Riffles | 2.95 (+/- 1.63) | 1.10 (+/- 0.683) |
| Runs | 2.37 (+/- 0.632) | 0.953 (+/- 0.352) |
| Pools | 2.02 (+/- 0.4.22) | 0.780 (+/- 0.180) |

Table 3. The mean (+/- 1 SD) Inverse Simpson Functional Diversity and mean Shannon Functional Diversity of each stream and of each habitat type.

| | *Inverse Simpson* | *Shannon* |
|---|---|---|
| Gross Cr | 1.86 (+/- 0.45) | 0.640 (+/- 0.221) |
| Hudson's Br. | 2.62 (+/- 0.804) | 1.04 (+/- 0.279) |
| Riffles | 2.45 (+/- 1.17) | 0.906 (+/- 0.514) |
| Runs | 2.26 (+/- 0.427) | 0.862 (+/- 0.213) |
| Pools | 2.00 (+/- 0.383) | 0.759 (+/- 0.163) |

Table 4. The mean (+/- 1 SD), median, minimum, and maximum unweighted tolerance values of each stream and of each habitat type.

| | *Site* | | *Habitat* | | |
|---|---|---|---|---|---|
| | Gross | Hudson's | Riffle | Run | Pool |
| Mean (+/- SD) | 7.63 (+/- 1.11) | 4.95 (+/- 2.39) | 5.44 (+/- 2.30) | 5.47 (+/- 2.39) | 6.18 (+/- 2.48) |
| Median | 8 | 5 | 6 | 5.5 | 8 |
| Minimum | 5 | 0 | 1 | 1 | 0 |
| Maximum | 8 | 8 | 8 | 8 | 8 |

Table 5. The mean (+/- 1 SD), median, minimum, and maximum abundance-weighted tolerance values of each stream and of each habitat type.

| | *Site* | | *Habitat* | | |
|---|---|---|---|---|---|
| | Gross | Hudson's | Riffle | Run | Pool |
| Mean (+/- SD) | 7.96 (+/- 0.104) | 7.21 (+/- 0.624) | 7.22 (+/- 0.814) | 7.70 (+/- 0.423) | 7.83 (+/- 0.271) |
| Median | 7.99 | 7.45 | 7.4 | 7.9 | 7.92 |
| Minimum | 7.68 | 6.03 | 6.03 | 6.94 | 7.45 |
| Maximum | 8 | 7.86 | 8 | 8 | 8 |

Table 6. The mean and standard deviation (Stand. Dev.) of the results of the model of organic matter consumption for the functional feeding groups in each stream. "Density" is the density of macroinvertebrates. "Consumption" is the modeled organic matter consumption for each functional feeding group.

| | | *Density (ind. $m^{-2}$)* | | *Consumption (g AFDM $m^{-2}$ $y^{-1}$)* | |
|---|---|---|---|---|---|
| Function | Site | Mean | Stand. Dev. | Mean | Stand. Dev. |
| Collector | Gross | 1053 | 966 | 11.3 | 10.4 |
| | Hudson's | 598 | 356 | 6.43 | 3.83 |
| Filterer | Gross | 7.77 | 23.3 | 0.193 | 0.58 |
| | Hudson's | 118 | 230 | 2.93 | 5.73 |
| Predator | Gross | 1252 | 887 | 3.68 | 2.61 |
| | Hudson's | 726 | 706 | 2.14 | 2.08 |
| Scraper | Gross | 0 | 0 | 0 | 0 |
| | Hudson's | 53.1 | 103 | 0.856 | 1.66 |
| Shredder | Gross | 5.18 | 8.74 | 1.25 | 2.04 |
| | Hudson's | 146 | 165 | 35.3 | 39.9 |

Table 7. The mean and standard deviation (Stand. Dev.) of the results of the model of organic matter consumption (g AFDM $m^{-2}$ $y^{-1}$) for the functional feeding groups in each habitat type. "Density" is the density of macroinvertebrates. "Consumption" is the modeled organic matter consumption for each functional feeding group

| | Function | Habitat | Mean | Stand. Dev |
|---|---|---|---|---|
| | | Pool | 14.53 | 5.18 |
| | Collector | Riffle | 1.46 | 0.38 |
| | | Run | 17.95 | 13.31 |
| | | Pool | 0.00 | 0.00 |
| | Filterer | Riffle | 0.58 | 1.00 |
| | | Run | 0.00 | 0.00 |
| | | Pool | 3.15 | 1.01 |
| Gross | Predator | Riffle | 3.59 | 3.67 |
| | | Run | 4.31 | 3.42 |
| | | Pool | 0.00 | 0.00 |
| | Scraper | Riffle | 0.00 | 0.00 |
| | | Run | 0.00 | 0.00 |
| | | Pool | 0.94 | 1.62 |
| | Shredder | Riffle | 0.94 | 1.62 |
| | | Run | 1.87 | 3.25 |
| | | Pool | 5.38 | 0.95 |
| | Collector | Riffle | 6.05 | 5.72 |
| | | Run | 7.85 | 4.50 |
| | | Pool | 0.58 | 0.58 |
| | Filterer | Riffle | 7.83 | 8.74 |

| | | | | |
|---|---|---|---|---|
| | | Run | 0.39 | 0.67 |
| | | Pool | 2.85 | 3.45 |
| Hudson's | Predator | Riffle | 1.36 | 1.01 |
| | | Run | 2.21 | 1.63 |
| | | Pool | 0.00 | 0.00 |
| | Scraper | Riffle | 1.88 | 2.77 |
| | | Run | 0.69 | 0.76 |
| | | Pool | 18.74 | 27.73 |
| | Shredder | Riffle | 66.53 | 57.63 |
| | | Run | 20.61 | 8.59 |

Table 8. Environmental description of each Stream and Habitat, where Mean Width is the mean (+/- 1 SD) measurement of the wetted surface of the stream from each of the 3 habitat replicates in each stream, Mean Depth is the mean (+/- 1 SD) of 3 measurements from each replicate habitat, the range of right bank heights and left bank heights of each habitat, high water mark range at each habitat, and canopy coverage of each habitat. The surrounding vegetation was called the dominant type of vegetation found. Sediment types included sand, pebble, bedrock, and cobble. Sand was classified as any sediment > 1 mm in diameter, pebble was classified as any sediment between 1 mm-1 cm in diameter, cobble was classified as any sediment between 1 cm-10 cm in diameter, and bedrock was classified as immovable rock substrate.

| Site | Habitat | Mean Width (m) | Mean Depth (cm) | R Bank range (m) | L Bank range (m) | HWM Range (m) | Sed | Veg | Canopy |
|---|---|---|---|---|---|---|---|---|---|
| Gross | Riffle | 2.23 (+/-.68) | 7.56 (+/- 4.44) | 2 to 3 | 1 to 1.5 | 3.5 | sand and cobble | trees, shrubs, grass | 0 to 24 |
| Gross | Pool | 3.7 (+/- 1.3) | 41.67 (+/- 22.56) | 2.5 to 3 | 1.5 to 2 | 3.5 | sand | trees, shrubs, grass | 24 |
| Gross | Run | 3.23 (+/- .40) | 14 (+/- 6.38) | 2.5 to 3 | 1 | 3.5 | sand and cobble | trees, shrubs, grass | 24 |
| Hudson | Riffle | 2.63 (+/- 1.19) | 4.22 (+/- .84) | 0.2 to 1.5 | 0.2 to 1.7 | 1 | cobble and bedrock | forest | 24 |
| Hudson | Pool | 1.7 (+/-.2) | 33.89 (+/- 14.95) | 0 to 1.5 | 0 to 1.3 | 1 | sand and cobble | forest | 24 |
| Hudson | Run | 2 (+/- .75) | 9.67 (+/- 3) | 0 to 1.5 | 0.3 to 2 | 1 | pebble sand and cobble | forest | 24 |

## Appendix

Appendix 1. The order, family, genus, and genus tag of each taxon along with their respective tolerance levels, functions, and number of individuals found as well as the site, habitat, and replicate number they were found in. 10.6084/m9.figshare.28840337

## Literature Cited

Anderson, R. V., & Day, D. M. (1986). Predictive quality of macroinvertebrate—habitat associations in lower navigation pools of the Mississippi River. *Hydrobiologia*, 136, 101-112.

Barnum, T. R., Weller, D. E., & Williams, M. (2017). Urbanization reduces and homogenizes trait diversity in stream macroinvertebrate communities. *Ecological Applications*, 27(8), 2428-2442.

Bressler, D. W., Stribling, J. B., Paul, M. J., & Hicks, M. B. (2006). Stressor tolerance values for benthic macroinvertebrates in Mississippi. *Hydrobiologia*, 573, 155-172.

Chadwick, M. A., Dobberfuhl, D. R., Benke, A. C., Huryn, A. D., Suberkropp, K., & Thiele, J. E. (2006). Urbanization affects stream ecosystem function by altering hydrology, chemistry, and biotic richness. *Ecological Applications*, *16*(5), 1796-1807.

Chessman, B. C., & McEvoy, P. K. (2012). Insights into human impacts on streams from tolerance profiles of macroinvertebrate assemblages. *Water, Air, & Soil Pollution*, *223*, 1343-1352.

Clarke, K. R. (1993). Non-parametric multivariate analyses of changes in community structure. *Australian Journal of Ecology*, *18*(1), 117-143.

Dance, K. W., & Hynes, H. B. N. (1980). Some effects of agricultural land use on stream insect communities. Environmental Pollution Series A, *Ecological and Biological*, 22(1), 19-28.

Ferreira, S. J. F., Pinel, S., Ríos-Villamizar, E. A., Miranda, S. Á. F., Pascoaloto, D., Vital, A. R. T., ... & da Cunha, H. B. (2021). Impact of rapid urbanization on stream water quality in the Brazilian Amazon. *Environmental Earth Sciences*, *80*, 1-16.

Ferrington Jr, L. C., Kavanaugh, R. G., Schmidt, F. J., & Kavanaugh, J. L. (1995). Habitat separation among Chironomidae (Diptera) in big springs. *Journal of the Kansas Entomological Society*, 152-165.

Fisher, S. G., & Gray, L. J. (1983). Secondary production and organic matter processing by collector macroinvetebrates in a desert stream. *Ecology*, 64(5), 1217-1224.

Hill, M. O. (1973). Diversity and evenness: a unifying notation and its consequences. *Ecology*, *54*(2), 427-432.

Hilsenhoff, W. L. (1987). An improved biotic index of organic stream pollution. *The Great Lakes Entomologist*, *20*(1), 7.

Gao, J., Peng, Z., Zang, H., Wang, Y., Ding, N., He, S., ... & Wang, B. (2023). Species sensitivity and functional uniqueness determine the response of macroinvertebrate functional diversity to species loss in urban streams. *Freshwater Biology*, 68(4), 674-688.

Graça, Cressa, Gessner, Feio, Callies, & Barrios. (2001). Food quality, feeding preferences, survival and growth of shredders from temperate and tropical streams. *Freshwater Biology*, 46(7), 947-957.

Greeves, S. (2023). Tracking Trash: Understanding Patterns of Debris Pollution in Knoxville's Urban Streams. Sustainability, 15(24), 16747.

Guo, F., Bunn, S. E., Brett, M. T., Fry, B., Hager, H., Ouyang, X., & Kainz, M. J. (2018). Feeding strategies for the acquisition of high-quality food sources in stream macroinvertebrates: Collecting, integrating, and mixed feeding. *Limnology and Oceanography*, *63*(5), 1964-1978.

Keinath, S., Onandia, G., Griesbaum, F., & Rödel, M. O. (2023). Effects of urbanization, biotic and abiotic factors on aquatic insect diversity in urban ponds. *Frontiers in Ecology and Evolution*, *11*, 1121400.

Komínková, D. (2012). The urban stream syndrome–a mini-review. *The Open Environmental & Biological Monitoring Journal*, *5*(1), 24-29.

Lenat, D. R. (1993). A biotic index for the southeastern United States: derivation and list of tolerance values, with criteria for assigning water-quality ratings. *Journal of the North American Benthological Society*, *12*(3), 279-290.

Macroinvertebrates.org. (n.d.). Atlas of Common Freshwater Macroinvertebrates of Eastern North America. https://www.macroinvertebrates.org

Merritt, R. W., Cummins, K. W., & Berg, M. B. (2017). Trophic relationships of macroinvertebrates. In *Methods in Stream Ecology, Volume 1* (pp. 413-433). Academic Press.

Meyer, J. L., Paul, M. J., & Taulbee, W. K. (2005). Stream ecosystem function in urbanizing landscapes. *Journal of the North American Benthological Society*, *24*(3), 602-612.

Monaghan, M. T., Thomas, S. A., Minshall, G. W., Newbold, J. D., & Cushing, C. E. (2001). The influence of filter-feeding benthic macroinvertebrates on the transport and deposition of particulate organic matter and diatoms in two streams. *Limnology and Oceanography*, 46(5), 1091-1099.

Newman, R. M. (1991). Herbivory and detritivory on freshwater macrophytes by invertebrates: a review. *Journal of the North American Benthological Society*, *10*(2), 89-114.

Oester, R., dos Reis Oliveira, P. C., Moretti, M. S., Altermatt, F., & Bruder, A. (2023). Leaf-associated macroinvertebrate assemblage and leaf litter breakdown in headwater streams depend on local riparian vegetation. *Hydrobiologia*, 850(15), 3359-3374.

Peng, X., Cheng, X., Zhu, D., & Huang, D. (2024). Application of functional feeding groups of macroinvertebrates in highly urbanized streams. *AQUA—Water Infrastructure, Ecosystems and Society*, 73(2), 338-354.

Poff, N. L., Olden, J. D., Vieira, N. K., Finn, D. S., Simmons, M. P., & Kondratieff, B. C. (2006). Functional trait niches of North American lotic insects: traits-based ecological

applications in light of phylogenetic relationships. *Journal of the North American Benthological Society*, 25(4), 730-755.

R Core Team (2024). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.

Reisinger, A. J., Doody, T. R., Groffman, P. M., Kaushal, S. S., & Rosi, E. J. (2019). Seeing the light: urban stream restoration affects stream metabolism and nitrate uptake via changes in canopy cover. *Ecological Applications*, 29(6), e01941.

Ross, D. H., & Wallace, J. B. (1983). Longitudinal patterns of production, food consumption, and seston utilization by net-spinning caddisflies (Trichoptera) in a southern Appalachian stream (USA). *Ecography*, *6*(3), 270-284.

Rosi-Marshall, E. J., & Wallace, J. B. (2002). Invertebrate food webs along a stream resource gradient. *Freshwater biology*, *47*(1), 129-141.

Schmid, P. E., & Schmid-Araya, J. M. (1997). Predation on meiobenthic assemblages: resource use of a tanypod guild (Chironomidae, Diptera) in a gravel stream. *Freshwater Biology*, 38(1), 67-91.

Speelman, J., & Carroll, N. (2016). *Bioindicators of Water Quality Quick–Reference Guide* [Review of *Bioindicators of Water Quality Quick–Reference Guide*]. Purdue Extension; Purdue University. https://ag.purdue.edu/department/asec/_docs/natural_resources/ace-11_bioindicators-2016.pdf

Stagliano, D. M., & Whiles, M. R. (2002). Macroinvertebrate production and trophic structure in a tallgrass prairie headwater stream. *Journal of the North American Benthological Society*, *21*(1), 97-113.

Sterling, J. L., Rosemond, A. D., & Wenger, S. J. (2016). Watershed urbanization affects macroinvertebrate community structure and reduces biomass through similar pathways in Piedmont streams, Georgia, USA. *Freshwater Science*, 35(2), 676-688.

Stoughton, S. J. (2017). Components of Urbanization and Urban Proximity Influence Water Quality in Low Gradient, Headwater Streams. University of Central Arkansas.

Sun, X., Chai, F. Y., Yu, H. X., Zhang, Y., Bao, X. X., Liu, M. H., & Wang, W. (2021). Impacts of environmental variables on macroinvertebrate functional feeding groups and biodiversity in a Muling River wetland from northeast China. *Applied Ecology & Environmental Research*, *19*(5).

Tampo, L., Kaboré, I., Alhassan, E. H., Ouéda, A., Bawa, L. M., & Djaneye-Boundjou, G. (2021). Benthic macroinvertebrates as ecological indicators: their sensitivity to the water quality and human disturbances in a tropical river. *Frontiers in Water*, *3*, 662765.

Tilman, D. (2001). Functional diversity. In *Encyclopedia of Biodiversity: Second Edition* (pp. 587-596). Elsevier Inc..

Townsend, C. R., & Hildrew, A. G. (1994). Species traits in relation to a habitat templet for river systems. *Freshwater biology*, *31*(3), 265-275.

Vannote, R. L., Minshall, G. W., Cummins, K. W., Sedell, J. R., & Cushing, C. E. (1980). The river continuum concept. *Canadian journal of fisheries and aquatic sciences*, 37(1), 130-137.

Violin, C. R., Cada, P., Sudduth, E. B., Hassett, B. A., Penrose, D. L., & Bernhardt, E. S. (2011). Effects of urbanization and urban stream restoration on the physical and biological structure of stream ecosystems. *Ecological Applications*, *21*(6), 1932-1949.

Violle, C., Navas, M. L., Vile, D., Kazakou, E., Fortunel, C., Hummel, I., & Garnier, E. (2007). Let the concept of trait be functional!. *Oikos*, *116*(5), 882-892.

Wallace, J. B., & Webster, J. R. (1996). The role of macroinvertebrates in stream ecosystem function. *Annual Review of Entomology*, *41*(1), 115-139.

Walsh, C. J., Roy, A. H., Feminella, J. W., Cottingham, P. D., Groffman, P. M., & Morgan, R. P. (2005). The urban stream syndrome: current knowledge and the search for a cure. *Journal of the North American Benthological Society*, *24*(3), 706-723.

Wan Abdul Ghani, W. M. H., Abas Kutty, A., Mahazar, M. A., Al-Shami, S. A., & Ab Hamid, S. (2018). Performance of biotic indices in comparison to chemical-based Water Quality Index (WQI) in evaluating the water quality of urban river. *Environmental Monitoring and assessment*, *190*, 1-14.